

Research Topic: Bio-Inspired cryptography

Source 1: <https://www.nature.com/articles/s41598-025-29168-y>

Source 2: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12118996/>

SLOs for Unit 6:

https://docs.google.com/document/d/1GyjiFy2fWG2ku8kZb_Ok2q6xrmQlqBaXevgodq0lAul/edit?usp=sharing

SLOs for Unit 3:

<https://docs.google.com/document/d/1qUS4hvxSEQFMdg2U88mFo8DtJWEyuO9nAVvFxSJEQKk/edit?usp=sharing>

Premise of how both relate: Bio-inspired cryptography uses biological processes and stochastic mechanisms to develop secure symmetric and asymmetric algorithms to improve computer security in a world threatened by quantum computer based brute force attacks

A. Research Topic:

For those of you true science lovers this is your opportunity to investigate a topic that we didn't cover or we didn't go into depth enough that it left you hanging. Are there current issues you want to discuss in science? Want to inform your peers about policy changes that may affect the planet? Are there problems you want to solve when you get out into the world? The end goal is to **create a physical product to display** for others to spark their interest in the subject you're passionate about.

Product ideas (although not limited to): brochure, poster, art piece, model, video

The end goal is to **create a physical product to display** for others to spark their interest in the subject you're passionate about.

Research topic ideas (although, not limited to):

1. Vaccination choice – pro- vs. anti vax
2. Human cloning
3. Altering human embryos (gene therapy)
4. Stem cell research (e.g. should embryos from fertility treatments be used to make new stem cell lines?)
5. Clinical trials (trying the safety of new medications on healthy and sick people)
6. End-of-life issues: withdrawal from active treatment; “Do-Not-Resuscitate” orders
7. Treatment choices in the face of terminal illnesses – prolonging life vs. palliative treatments; Choice of treatment – cost vs. effectiveness
8. Religious objections to medical treatment (e.g. no blood transfusion or healing with prayer)
9. Equity in access to treatment (medical insurance coverage; availability of modern medicine, etc.)
10. Medical insurance coverage
11. Relevant climate change legislation
12. ...and so much more!

****The product must fully explain the concept/topic/process and information must be accurate. You must also connect your research back to at least 5 of the Big Ideas, Enduring Understandings and/or SLOs as is relevant. [Link to Framework](#)**

All projects must include: Works Cited

*Note, for all project options, make sure you conduct relevant and accurate research. At minimum, you should create a Works Cited page in MLA/APA format with TWO credible sources. Depending on your topic, you may very well need more!

AP Biology EOC Interest Project - Final Rubric (Total: 16 Summative Points)

Category	4 Points (Exceeds Expectations)	3 Points (Meets Expectations)	2 Points (Approaches Expectations)	1 Point (Needs Improvement)
Physical Product & Presentation	Product is visually engaging, professionally presented, includes a clear title, 3+ visuals, and is thoughtfully organized for display.	Product is neat, includes a clear title and 2+ visuals, and is mostly organized for display.	Product is somewhat neat, includes a title and at least 1 visual, with some attempt at organization.	Product is disorganized or incomplete, with unclear visuals or missing required elements.
Content Mastery & Conceptual Depth	Demonstrates a thorough understanding of the topic; all required key concepts (as outlined per project option) are addressed in detail and with accuracy.	Demonstrates a good understanding of the topic; most key concepts are addressed accurately and in reasonable detail.	Demonstrates partial understanding; several key concepts are missing or underdeveloped.	Demonstrates limited understanding; few key concepts are addressed, and/or major inaccuracies are present.
Connection to AP Biology Framework	Option A: Strong and clear connection to 5 or more AP Biology Big Ideas, topics, SLOs, LTs with accurate, in-depth analysis. Option B: Strong connection to 10 or more AP Biology concepts (SLOs/LTs), with explicit explanation of how each applies to your selected organ system.	Connection to AP Biology is present (3-4) and mostly accurate, with some analysis or explanation. Connection to 7-9 or more concepts we learned that apply to your organ system.	Some attempts at connecting to AP Biology, but lack clarity, depth, or accuracy. 1-2 connections to AP Biology concepts and/or 3-6 connections to organ systems.	Few weak or missing connections to AP Biology framework and/or organ systems.

Research & Citations	Includes at least 2 credible sources properly cited in MLA or APA format; sources strongly support project content.	Includes 2 sources, mostly cited correctly, with appropriate relevance to the topic.	Includes 1–2 sources with citation errors or questionable credibility.	Missing citations or uses non-credible/unrelated sources.
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Omega deoxyribonucleic acid cryptography key-based authentication

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[Kamaruddin Malik Mohamad](#)

[Scientific Reports](#)

15, Article number: 45473 (2025) [Cite this article](#)

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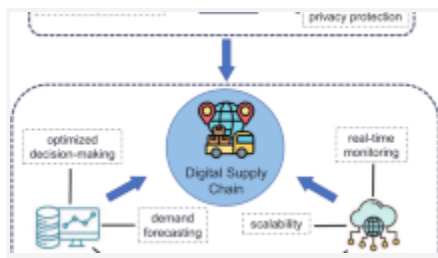
[Metrics](#)

Abstract

Deoxyribonucleic acid cryptography is a biologically inspired approach characterized by low computational complexity. It employs biological principles to create cryptographically strong ciphers, making it particularly suitable for protecting sensitive data on resource-constrained devices. However, the existing literature lacks solutions for securing authentication mechanisms tailored for these resource-constrained devices. To bridge this gap, the current study proposes a novel authentication design rooted in

deoxyribonucleic acid cryptography, namely omega deoxyribonucleic acid cryptography key-based authentication. The proposed omega deoxyribonucleic acid cryptography-based authentication method aligns with contemporary standards for cryptographic systems and delivers a security level quantified at 256 bits of complexity. To validate its resilience, one tests the collision resistance of the proposed authentication mechanism using the standard Dieharder statistical test suite, where the mechanism successfully passes the collision resistance test. Additionally, the proposed scheme is mathematically proven secure against existential forgery under a chosen message attack.

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Introduction

The technologies of resource-constrained computing devices have gained significant prominence in recent years^{13,21}. For example, sensor networks, smart cards, and the Internet of Things (IoT) are included in this category^{11,37}. The data generated by these devices needs to be protected in terms of confidentiality, and the origin of the transmitted data needs to be authenticated²⁴. These resource-constrained computing devices have limited capabilities in terms of processing power, memory, storage, and energy consumption. Lightweight cryptography provides a low-complexity solution to enhance security for these devices. One type of lightweight cryptography, known as deoxyribonucleic acid (DNA) cryptography, offers faster computing speeds while requiring minimal storage and low power consumption^{8,15}.

DNA cryptography represents the merging of cryptography and molecular biology, driven by advances in biotechnology and DNA computing. This integration has led to the emergence of DNA cryptography and information science, a domain pioneered by Adleman's groundbreaking work in DNA computing²⁶. Although this nascent field has rapidly attracted global interest, with numerous research projects underway, it remains in the early stages of development within cryptography, with most investigations focusing on foundational concepts¹⁶.

DNA cryptography utilizes biological principles to pioneer new computing paradigms. The concept of a DNA computer envisions matching the processing capabilities of traditional systems while addressing problems currently considered intractable. It exploits DNA's unique attributes: exceptional parallel processing capacity, which can perform over one billion operations per second and energy efficiency¹⁰. These features position DNA-based computing as a transformative frontier in secure information processing.

DNA cryptography explores the use of four types of nitrogenous bases in DNA. The nitrogenous bases are adenine (A), guanine (G), cytosine (C), and thymine (T)²⁰. The researchers form a pseudorandom cryptographically keys based on the nitrogenous bases. These keys are used for encoding, storing, and transmitting information securely. To date literature shown that the DNA cryptography is designed mainly to ensure data confidentiality while transmitting over insecure network. These design only can protect the data from being read by unauthorized parties but does not authenticate the sender. For example, encryption scheme using DNA technology⁹, DNA-based cryptographic framework³⁴, DNA security algorithm using DNA codons³⁰, DNA with Pi-based key generating encryption algorithm³⁶, DNA-Based S-Box¹, DNA-based Vernam Cipher³⁵ and Omega network pseudorandom key generation based on DNA cryptography (ONDNA)²⁹.

Authentication is a cryptographic security mechanism used to verify the identity of a message^{14,27}. This process ensures both the integrity and the source of messages during transmission. There are examples of hybrid cryptographic systems that integrate DNA cryptography and hash functions for authenticating messages during transmission. For instance, DNA-BRNG key and a novel hash algorithm³², as well as DNA-LCG key and a novel hash algorithm³³. Both designs propose their own pseudorandom number generators based on the nitrogenous bases in DNA. The output of the pseudorandom number generator is known as a pseudorandom cryptographic key. The message authentication code is generated by applying the novel hash algorithm to the message and the pseudorandom cryptographic key. A drawback is that these studies fail to propose a complete authentication process based on DNA cryptography.

Hence, this design addresses a significant gap in the existing literature, as prior research has not extensively explored DNA cryptography for key-based authentication. By filling this gap, this research introduces an encrypt-then-authenticate process that ensures both the confidentiality and authenticity of data¹⁷. The process involves two main steps: encryption

followed by authentication. The encryption process is rooted in the Omega network pseudorandom key generation (ONDNA), which produces ciphertext of the same length as the input message. The ciphertext is then authenticated, generating a fixed 256 bits tag. We designate this complete process as ODNAMAC.

It is important to highlight that the existing ONDNA is fundamentally based on DNA cryptography principles, tailored specifically to maximize memory efficiency. This design choice allows for rapid processing speeds and scalability, making it suitable for large-scale applications involving extensive data encryption and decryption tasks²⁹. Consequently, the tags generated by the ODNAMAC process inherit these advantageous properties, ensuring that they are not only produced quickly but also support scalable operations, thereby facilitating efficient and secure data authentication in cryptographic environments.

The key advantage of our proposed encrypt-then-authenticate² ODNAMAC approach lies in its verification optimization: when tag matching fails during verification, decryption becomes unnecessary because message integrity cannot be guaranteed. This approach minimizes the computational steps involved in decryption, optimizes the authentication process, and enhances overall efficiency by conserving computational resources¹².

We present a literature review in Sect. "Literature review". Section "Deoxyribonucleic acid" generally introduces deoxyribonucleic acid (DNA). Section "Omega network pseudorandom key generation based on DNA cryptography" discusses Omega network pseudorandom key generation based on DNA cryptography. Section "Message authentication Code" introduces the theoretical definition of message authentication codes. Section "Research Design" shows the research design. Section "The proposed ODNAMAC scheme" presents our proposed ODNAMAC scheme. Section "ODNAMAC analysis and discussion" provides the ODNAMAC analysis and discussion, including tag generation, statistical tests, formal security analysis,

brute-force attacks, and collision attacks. Section “Conclusion” concludes the paper.”

Literature review

The DNA-based encryption scheme is a cryptographic approach that combines biological DNA operations with computational methods to improve data security. It encodes plaintext into DNA nucleotides bases A, C, T, and G using binary mapping rules, utilizing eight pattern types (CTAG, CATG, GTAC, GATC, TCGA, TGCA, ACGT, AGCT) that are topologically equivalent and follow the complementary pairing rules of nucleotides⁹. The authors utilize Watson-Crick complementarity for diffusion and produce pseudorandom keys via DNA oligonucleotide synthesis. Their encryption method combines DNA substitution and permutation techniques inspired by polymerase chain reactions (PCR) to enhance diffusion and confusion. Even though the encryption and decryption processes are effective, the authors do not provide formal proofs or statistical analyses to substantiate the scheme’s security, leaving its robustness uncertain.

Sukumaran and Misbahuddin introduce a DNA-based cryptographic framework designed to improve cloud data security by encoding sensitive information into synthetic DNA sequences³⁴. This approach involves converting plaintext into binary, then mapping it to nucleotide bases through a deterministic algorithm, and utilizing PCR along with Watson-Crick complementarity to diffuse the data. Encryption keys are derived from unique synthetic DNA nucleotide sequences to ensure resistance against brute-force attacks. The scheme also incorporates error-correcting codes to mitigate the synthesis errors commonly associated with DNA storage. Their security analysis focuses primarily on two key aspects: integrity and confidentiality, while authentication was not addressed.

Patnala and Kumar present a hierarchical DNA encryption algorithm that utilizes codon-level encoding to improve data security³⁰. The process begins

by converting plaintext into ASCII codes, which are then transformed into binary. These binary sequences are mapped to DNA nucleotides using a triplet codon scheme, such as assigning 3 bits to each codon which facilitating multi-layered diffusion. The algorithm incorporates three tiers of security: base substitution through codon shuffling, position permutation, and stochastic noise injection, providing scalable protection suitable for DNA-based storage and biomedical communication. However, the authors only report the encryption time and do not include a security analysis.

The paper introduces a hybrid cryptographic approach that integrates DNA-based encoding with pi-derived key generation to improve security and randomness³⁶. Initially, the plaintext is transformed into a binary stream, which is then mapped to DNA nucleotides via a pi-based mapping scheme that prevents repetitive patterns. For key creation, the algorithm derives pseudorandom sequences from pi values and processes them using DNA diffusion synthesis rules to produce biologically inspired cryptographic keys with increased entropy. The encryption process utilizes substitution-permutation networks (SPNs) to achieve diffusion and confusion. However, as noted in⁶, this method is vulnerable to man-in-the-middle attacks. The authors suggest addressing this security flaw by implementing a key exchange protocol⁶.

The DNA-based S-Box technique is an example of a DNA cryptography cipher. The S-Box includes the SPN that manipulates the nitrogenous bases A, C, T, and G. The SPN is essential to enhance the pseudorandomness and complexity of the S-Box output. Hence, satisfies key cryptographic properties, such as nonlinearity, uniformity in differential analysis, and resistance to linear attacks¹.

The DNA-based Vernam Cipher is another example of a symmetric DNA cryptography cipher which integrates the security of the traditional Vernam Cipher with the complexity and pseudorandomness of DNA sequences³⁵. The

goal is to develop a highly secure encryption technique that utilizes the uniqueness characteristics of DNA. The process includes DNA encoding, key generation, encryption and decryption. The DNA encoding process, the plaintext message is converted into a predefined mapping binary format. Next, the key generation, a random DNA sequence is generated to serve as the one-time pad key. This key length is same length as the plaintext DNA sequence. As for encryption process, the DNA-based Vernam Cipher performs a XOR operation between the plaintext DNA sequence and the key DNA sequence. While, the decryption process is the reverse process of encryption, by performing the XOR operation between the ciphertext DNA sequence with the corresponding key DNA sequence³⁵.

The Omega network pseudorandom key generation based on DNA cryptography is characterized as a low-complexity cipher for generating pseudorandom keys²⁹. The Omega Network is used to create a complex and nonlinear transformation of input data. This transformation is essential for generating pseudorandom sequences as secret keys for encryption and decryption. The network's routing capabilities ensure that the output is highly dependent on the input, making it difficult for adversaries to predict the key.

Sodhi and Gaba introduced a MAC scheme that utilizes a DNA-based random number generator (DNA-BRNG) key to derive unique DNA sequences, generating highly unpredictable private keys and enhancing protection against key-guessing attacks³². For authentication purposes, they employ a conventional hash function in conjunction with this private key to improve efficiency and collision resistance, thereby ensuring reliable message authentication. By integrating DNA-generated randomness with a tailored hash function, their scheme attains robust security alongside practical computational performance. Sodhi et al. further extended their research by combining the DNA-based linear congruential generator (DNA-LCG) with a novel hash algorithm for message authentication³³. This method utilizes unique DNA sequences to generate highly unpredictable cryptographic keys, while the LCG further strengthens randomness. A new hash algorithm is introduced to improve collision resistance and efficiency in MAC computation. By integrating biological data with traditional pseudorandom number

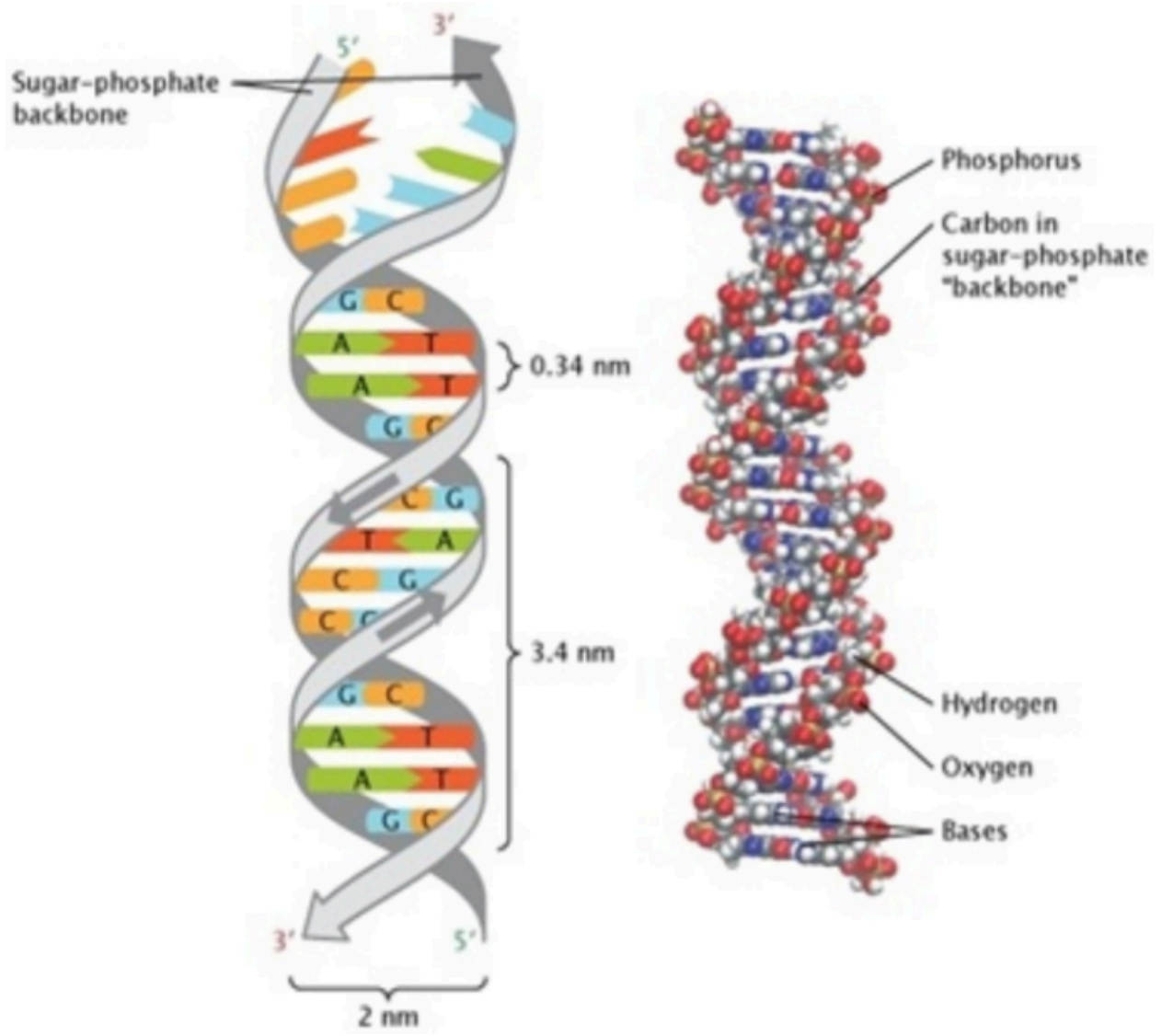
generation, the scheme aims to resist brute-force and key-guessing attacks. Shaw, et. al. proposes an integration genomic data together with traditional hash function as keyed-hash message authentication code (HMAC) protocol³¹. The approach involves extracting genomic features, processing them into secure keys, and integrating them with traditional HMAC mechanisms. By utilizing the inherent variability and complexity of DNA sequences, the protocol aims to improve resistance against brute-force and forgery attacks. A drawback is that these studies fail to propose a complete authentication process based on DNA cryptography.

Deoxyribonucleic acid

Deoxyribonucleic Acid (DNA) functions as the molecular blueprint of life, controlling organism development and biological activities. This remarkable macromolecule contains all genetic information within an organism and is located in the nucleus of each cell, earning its status as a fundamental component of life²⁵. As the universal genetic material in all cellular life forms, DNA orchestrates cellular processes such as growth, reproduction, and programmed cell death through its complex chemical structure.

The well-known double-helix structure of DNA, initially described by Watson and Crick, consists of two antiparallel strands that twist into a ladder-like formation (²⁸; see Fig. 1). This long polymer is made up of millions of nucleotides arranged in specific sequences, with genetic information encoded by different combinations of four nitrogenous bases³. According in²⁸, the variations in these base sequences generate genetic diversity and underpin DNA's unique replication process. During semiconservative replication, the double helix unwinds, and each strand serves as a template for synthesizing a complementary strand, resulting in two daughter molecules with each containing one original and one newly formed strand.

Fig. 1



[Full size image](#)

DNA double helix structure²⁸.

The structural intricacy of DNA allows for nearly limitless genetic variation. With about one million base pairs in a typical DNA molecule and billions of potential nucleotide arrangements within the double helix, DNA provides the molecular foundation for biological diversity. This sophisticated system of information storage and transfer is fundamental to all known life processes.

Omega network pseudorandom key generation based on DNA cryptography

The Omega network pseudorandom key generation based on DNA cryptography (ONDNA), as shown in Algorithm 1, consists of three rows and four columns, denoted as ²⁹. It incorporates four operations or DNA rules: DNA XOR rules (represented by), mRNA mapping rules where the first letter precedes the second (denoted as), mRNA mapping rules where the second letter precedes the first (denoted as), and complementary mapping rules (denoted as). The inputs are two 128 bits of private strings, namely sperm and ovum. The output is 256 bits of cryptographically pseudorandom key. Specifically, the elements , , and are generated using DNA XOR rules. Meanwhile, , , and follow mRNA mapping rules with the first letter leading the second. , , and adhere to mRNA mapping rules where the second letter comes before the first. Lastly, , , and are derived from complementary mapping rules. These operations are randomly arranged to ensure an even distribution across the design, enabling the generation of a cryptographically pseudorandom key-based on DNA. These cryptographically pseudorandom keys are used for encryption and decryption to ensure the confidentiality of data during the transmission over insecure internet.

In one round of the ONDNA key generation, a 128 bits sperm and a 128 bits ovum are used to generate a 256 bits cryptographic pseudorandom key-based DNA. If an application or protocol demands more than 256 bits, the process is repeated. For instance, to obtain 400 bits of pseudorandom key-based DNA, two iterations are needed. The first round, as outlined, produces a 256 bits pseudorandom key-based DNA. The last round, the initial 128 bits (0127) and the subsequent 128 bits (128255) of the first output serve as the sperm and ovum, respectively. The key generation process is repeated (steps 1 through 11). Together, these two rounds yield 512 bits of pseudorandom key-based DNA. Since the application requires only 400 bits, the final 112 bits are discarded.

Algorithm 1

Require: Inputs: Strings of sperm (0...127) and string of ovum (0...127)

Require: Variable strings: x (0...127), y (0...127), z (0...127); $x, y, z \in \text{Pi value}$; zygote (0...127); $B_{i,j}$ (0...127); $i, j \in \{1, 2, 3, 4\}$

1. $\text{zygote} \leftarrow \text{sperm} \oplus \text{ovum}$
2. $x \leftarrow \text{Rand}(128)$
3. $y \leftarrow \text{Rand}(128)$
4. $z \leftarrow \text{Rand}(128)$
5. $B_{1,1}, B_{1,2}, B_{1,3}, B_{1,4} \leftarrow \text{zygote} \oplus y, x \oplus z, x \oplus y, \text{zygote} \oplus z$
6. $B_{1,1}, B_{1,2}, B_{1,3}, B_{1,4} \leftarrow \sigma(B_{1,1}), \mu(B_{1,2}), \#(B_{1,3}), \mu(B_{1,4})$
7. $B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4} \leftarrow B_{1,1} \oplus B_{1,3}, B_{1,1} \oplus B_{1,2}, B_{1,4} \oplus B_{1,2}, B_{1,4} \oplus B_{1,3}$
8. $B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4} \leftarrow \sigma(B_{2,1}), \#(B_{2,2}), *(B_{2,3}), *(B_{2,4})$
9. $B_{3,1}, B_{3,2}, B_{3,3}, B_{3,4} \leftarrow B_{2,1} \oplus B_{2,4}, B_{2,1} \oplus B_{2,3}, B_{2,3} \oplus B_{2,2}, B_{2,4} \oplus B_{2,2}$
10. $B_{3,1}, B_{3,2}, B_{3,3}, B_{3,4} \leftarrow \#(B_{3,1}), \sigma(B_{3,2}), \mu(B_{3,3}), *(B_{3,4})$
11. $\text{key} \leftarrow B_{3,1} \oplus B_{3,4} \parallel B_{3,2} \oplus B_{3,3}$

Output: String: key (0...255)

[Full size image](#)

ONDNA²⁹

The DNA XOR rules, as presented in Table [1](#), consist of combinations involving A, C, G, and T. In this encoding, A corresponds to 10, C to 01, G to 11, and T to 00. There are four rules, and the rule applied depends on the first character of the string. For instance, consider the first string, TCGA, which translates to the binary value 00011110, and the second string, AACT, which translates to the binary value 10100100. Because the first string begins with 'T', the output follows the first rule of DNA XOR. As a result, the XOR operation yields CGTT, with a binary value of 01110000.

Table 1 DNA XOR rules ^{[29](#)}.

[Full size table](#)

Table [2](#) presents the mRNA mapping rules. For instance, take the string CGTT, which encodes the binary value 01110000. The first letter and second letter are C and G, respectively; following the mRNA mapping rules, this yields AT, represented by the binary value 1000. Next, the third and fourth letters are T, producing GA according to the mRNA rules, which corresponds to the binary

value 1110. Therefore, the result is ATGA, with a binary representation of 10001110.

Table 2 mRNA mapping rules [29](#).

[Full size table](#)

Table [3](#) illustrates the complementary mapping of nucleotide letters in binary format. Four binary values represent the base of each nucleotide letter. For instance, consider the binary sequence 1010, which corresponds to the bases C and T. The complementary result is 1001. In another example, with the input string 10001110, the complementary output is 01111011.

Table 3 Complementary Mapping Rules [29](#).

[Full size table](#)

Definition 1

ONDNA is defined as:

- *sperm* is private string, it is chosen from the public space , such that with the length (128 bits) and probability distribution .
- *ovum* is private string, it is chosen from the public string space , such that with the length (128 bits) and probability distribution .
- *x* is public random string consisting of space , such that , with the length (128 bits) and probability distribution *X*.
- *y* is public random string consisting of space , such that , with the length (128 bits) and probability distribution *Y*.
- *z* is public random string consisting of space , such that , with the length (128 bits) and probability distribution *Z*.
- *K* is a 256 bits pseudorandom cryptographic key.

The output of ONDNA is 256 bits block of pseudorandom cryptographic keys which follows the Definition [2](#). This algorithm generates cryptographic keys of arbitrary length by producing multiple 256-bit blocks and discarding any excess bits.

Definition 2

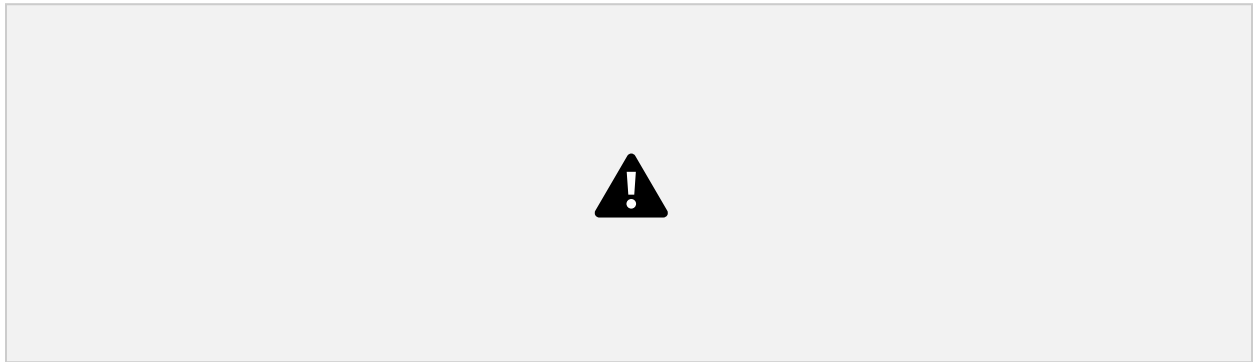
A pseudorandom keystream generator is considered secure only if no polynomial time algorithm can distinguish between the pseudorandom string generated by the pseudorandom keystream generator and a truly random string of the same length with a probability significantly greater than [19](#).

Message authentication code

The message authentication code (MAC) is a symmetric key cryptographic algorithm that ensures parties with the shared secret key may communicate securely and identify any unauthorized modifications to the message during transmission^{[19,23](#)}. MAC generates a tag that verifies the authenticity of a message. This technique produces fixed-length tags for messages of arbitrary

length. Figure 2 shows the general model of MAC. Both the sender and receiver use the same secret key to perform the message authentication process. The receiver generates a tag and compares it with the received tag. If both tags match, the message is regarded as authentic; if not, it is considered to have been tampered with by an unauthorized entity.

Fig. 2



[Full size image](#)

Message authentication code general model.

Definition 3

Message authentication code (MAC) over the message is defined as:

$\text{MAC}(\text{Gen}, \text{Tag}, \text{Ver})$ ¹⁹.

- Gen is a pseudorandom polynomial time algorithm that generates a key, such that .
- Tag is a pseudorandom polynomial time algorithm that on input key k and message m output a tag, .
- Ver is a deterministic polynomial time algorithm that on input key k , message m and tag t , output , where represents “accept”, represents “reject”.
- For all and , $\text{Pr}[= \text{“accept”}] = 1$.

The MAC scheme must secure existentially unforgeable under chosen message attacks where an attacker is able to generate at least one valid message and tag pair without possessing the secret key with negligible probability. This property guarantees that the MAC scheme provides strong integrity and authenticity assurances, preventing attackers from forging valid tags without knowledge of the secret key.

Definition 4

A MAC scheme $\text{MAC}(\text{Gen}, \text{Tag}, \text{Ver})$ is said to be existentially unforgeable under chosen message attacks if, for any generated key k , a polynomial time adversary can make tag and verification queries for arbitrary messages using k , to forge a valid tag t for a message m , the following advantage is negligible. $\text{Pr}[\dots]$ ⁴.

The MAC must prevent an attacker from finding two distinct messages m_1 and m_2 , that produce the same tag, $T(m_1) = T(m_2)$, where $m_1 \neq m_2$. This property is known as collision resistance of the MAC. Ensuring this property prevents collision attacks, where an attacker attempts to find two different messages that produce the same MAC, which could compromise the integrity and authenticity assurances provided by the MAC.

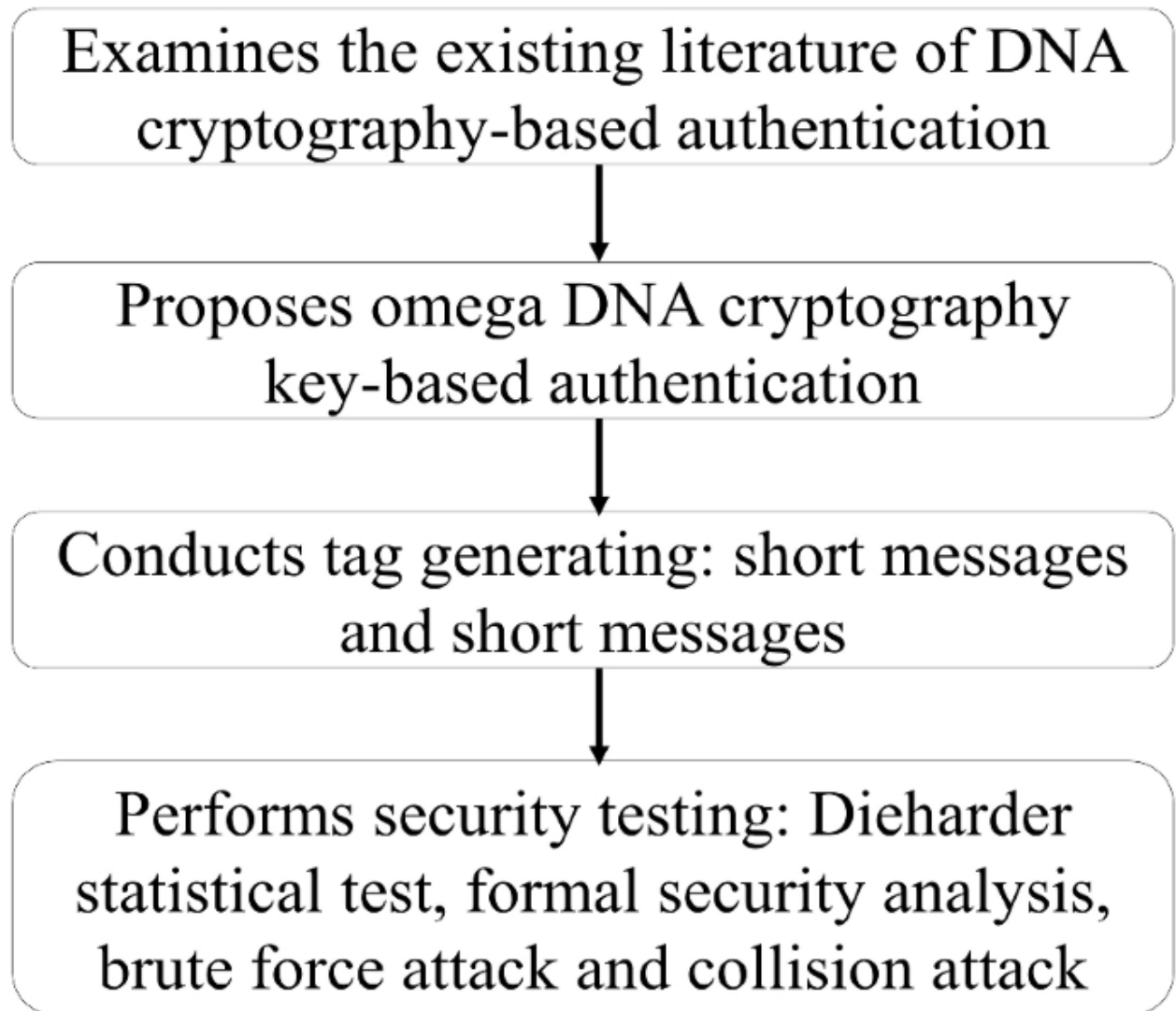
Definition 5

Let a MAC has a tag generation function, $Tag(k, m)$. A collision for Tag as a pair (m_1, m_2) such that $Tag(k, m_1) = Tag(k, m_2)$ and $m_1 \neq m_2$. The tag generation function Tag is called collision-resistant, if for any polynomial time adversary, the probability that the adversary output the collision is negligible⁵.

Research design

Figure 3 illustrates the research design, which consists of four major processes. The first step is to examine the existing literature on DNA cryptography-based authentication. Next, the study proposes an omega DNA cryptography key-based authentication method. This is followed by a phase of conducting tag generation. Finally, the research performs security testing on the proposed method.

Fig. 3



[Full size image](#)

Research design.

Firstly, this research provides a comprehensive examination of the literature on DNA cryptography as it applies to authentication. By exploring existing work on authentication protocols, MAC, and security vulnerabilities, this study identifies a critical gap in the field, where the current literature lacks fully developed DNA cryptography-based methods for message authentication.

Next, an omega DNA cryptography-based authentication method is proposed, comprising three algorithms for key generation, signing, and verification. The key generation algorithm, which uses an Omega network pseudorandom key generator, creates a 256 bits tag to ensure the confidentiality of the message.

Third, the proposed omega DNA cryptography-based authentication is implemented in C++ to support the tag generation and security evaluation phases. Four tags are generated: two use long messages that differ by only a single full stop, demonstrating one bit sensitivity, and two use short messages to confirm that the tag is fixed at 256 bits.

The security test includes the Dieharder statistical test, a formal security analysis, a brute force attack, and a collision attack. The Dieharder statistical test confirms the tag's unpredictability and security against the birthday paradox. The formal security analysis uses mathematical models and logic to prove that the proposed omega DNA cryptography key-based authentication is resilient against existential forgery attacks. A brute force attack systematically checks every possible key combination until it finds the correct one. The proposal generates a 256 bits tag, and the brute force test demonstrates that the key space for this tag is large enough to resist an exhaustive search, confirming its strength against this type of attack. For the collision attack, this research uses the birthday paradox to calculate the complexity of finding two different messages that produce the same 256 bits authentication tag. This experiment demonstrates that such an attack is computationally infeasible, proving the authentication method's security.

The proposed ODNAMAC scheme

The Omega DNA cryptography key-based authentication (ODNAMAC) is designed to provide message confidentiality and authenticate the message as shown in Algorithm 1 and Fig. 4. If the message fails authentication, the message may be being altered during transmission. For confidentiality, the message is encrypted using the ONDNA process. For authentication, the design consists of three algorithms: a key generation algorithm, a signing algorithm, and a verification algorithm. The key generator for authentication is ONDNA. The signing algorithm uses DNA XOR rules (denoted by the symbol $\oplus$, refer to Table 1), mRNA mapping rules (denoted by the symbol $\rightarrow$, refer to Table 2), and complementary mapping rules (denoted by the symbol $\sim$, refer to Table 3). The use of these rules aims to make it computationally infeasible for an attacker to generate a valid tag for a message without knowing the secret key. It should be noted that the message is of arbitrary length, and therefore, the ciphertext is also of arbitrary same length as message. However, the signing algorithm generates a fixed-length tag of 256 bits. The verification algorithm is used to compare the generated tag with the received tag.

Algorithm 2

Require: Inputs: Strings of sperm ($0 \dots 127$), string of ovum ($0 \dots 127$) and message m

Require: Variable strings: x ($0 \dots 127$), y ($0 \dots 127$), z ($0 \dots 127$); $x, y, z \in \text{Pi value}$; zygote ($0 \dots 127$); $B_{i,j}$ ($0 \dots 127$): $i, j \in \{1, 2, 3, 4\}$

Require: Registers: r_o ($0 \dots 255$), r_p ($0 \dots 255$), r_q ($0 \dots 255$) **Require Keys:** Encryption key k_e , authentication key k_a .

Encryption process:

1. zygote $\leftarrow$ sperm $\oplus$ ovum
2. $x \leftarrow \text{Rand}(128)$
3. $y \leftarrow \text{Rand}(128)$
4. $z \leftarrow \text{Rand}(128)$
5. $B_{1,1}, B_{1,2}, B_{1,3}, B_{1,4} \leftarrow \text{zygote} \oplus y, x \oplus z, x \oplus y, \text{zygote} \oplus z$
6. $B_{1,1}, B_{1,2}, B_{1,3}, B_{1,4} \leftarrow \sigma(B_{1,1}), \mu(B_{1,2}), \#(B_{1,3}), \mu(B_{1,4})$
7. $B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4} \leftarrow B_{1,1} \oplus B_{1,3}, B_{1,1} \oplus B_{1,2}, B_{1,4} \oplus B_{1,2}, B_{1,4} \oplus B_{1,3}$
8. $B_{2,1}, B_{2,2}, B_{2,3}, B_{2,4} \leftarrow \sigma(B_{2,1}), \#(B_{2,2}), *(B_{2,3}), *(B_{2,4})$
9. $B_{3,1}, B_{3,2}, B_{3,3}, B_{3,4} \leftarrow B_{2,1} \oplus B_{2,4}, B_{2,1} \oplus B_{2,3}, B_{2,3} \oplus B_{2,2}, B_{2,4} \oplus B_{2,2}$
10. $B_{3,1}, B_{3,2}, B_{3,3}, B_{3,4} \leftarrow \#(B_{2,1}), \sigma(B_{2,2}), \mu(B_{2,3}), *(B_{2,4})$
11. key $\leftarrow B_{3,1} \oplus B_{3,4} \parallel B_{3,2} \oplus B_{3,3}$
12. Return encryption key k_e .
13. $c \leftarrow \text{Encryption}(k_e \oplus m)$

Authentication process:

Step I: Key generation, Gen. Steps 1 through 11 are repeated to generate 256 bits authentication key k_a .

Step II: Tag generation, Tag. c is divided into blocks, such that c_i , where $i \in \{1, 2, \dots, L\}$. Each block is 256 bits. If last block is not 256 bits, padding is necessary with 0 to make the total block length 256 bits.

r_q is initiated with 256 bits of 0 value.

for $i \leftarrow 1$ to L **do**

if $i = 1$ to $L - 1$ **then**

$r_o \leftarrow *(k_a, c_i)$

$r_p \leftarrow \#(r_o)$

$r_q \leftarrow *(r_p)$

else

$t \leftarrow \sigma(r_{qL})$

end if

end for

String $c \parallel t$ is sent to receiver. Receiver receives $c \parallel t$. Receiver decrypts the c to get the m .

Step III: Tag verification, Ver. Step II is repeated to generate the t' .

if $t == t'$ **then**

 Return 1.

else

 Return 0.

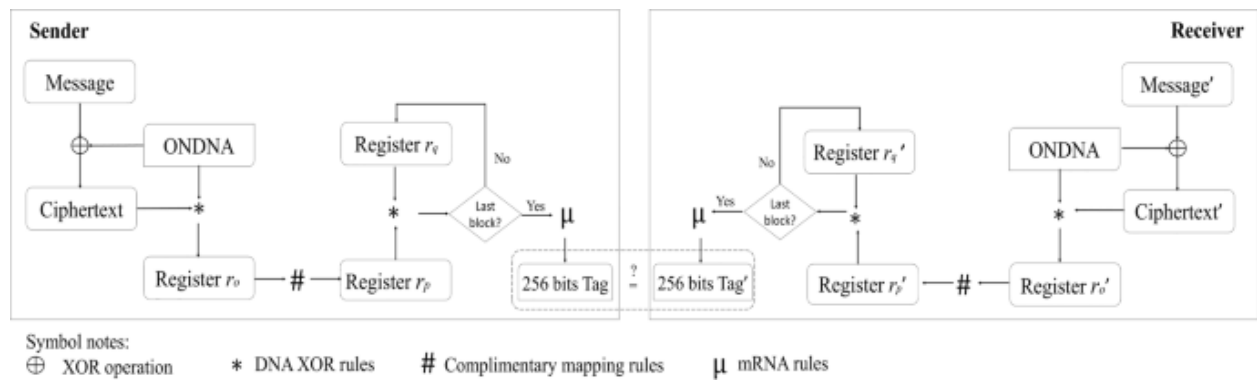
end if

Output: String: message m , ciphertext c , tag t

[Full size image](#)

ODNAMAC

Fig. 4



[Full size image](#)

ODNAMAC general flowchart.

Definition 6

A ODNAMAC(Gen,Tag,Ver) is existential forgery under a chosen message attack secure, if no polynomial time adversary can produce a valid forgery message (m, t) for a new ciphertext c , who can win the ODNAMAC oracle game with probability not larger than ϵ .

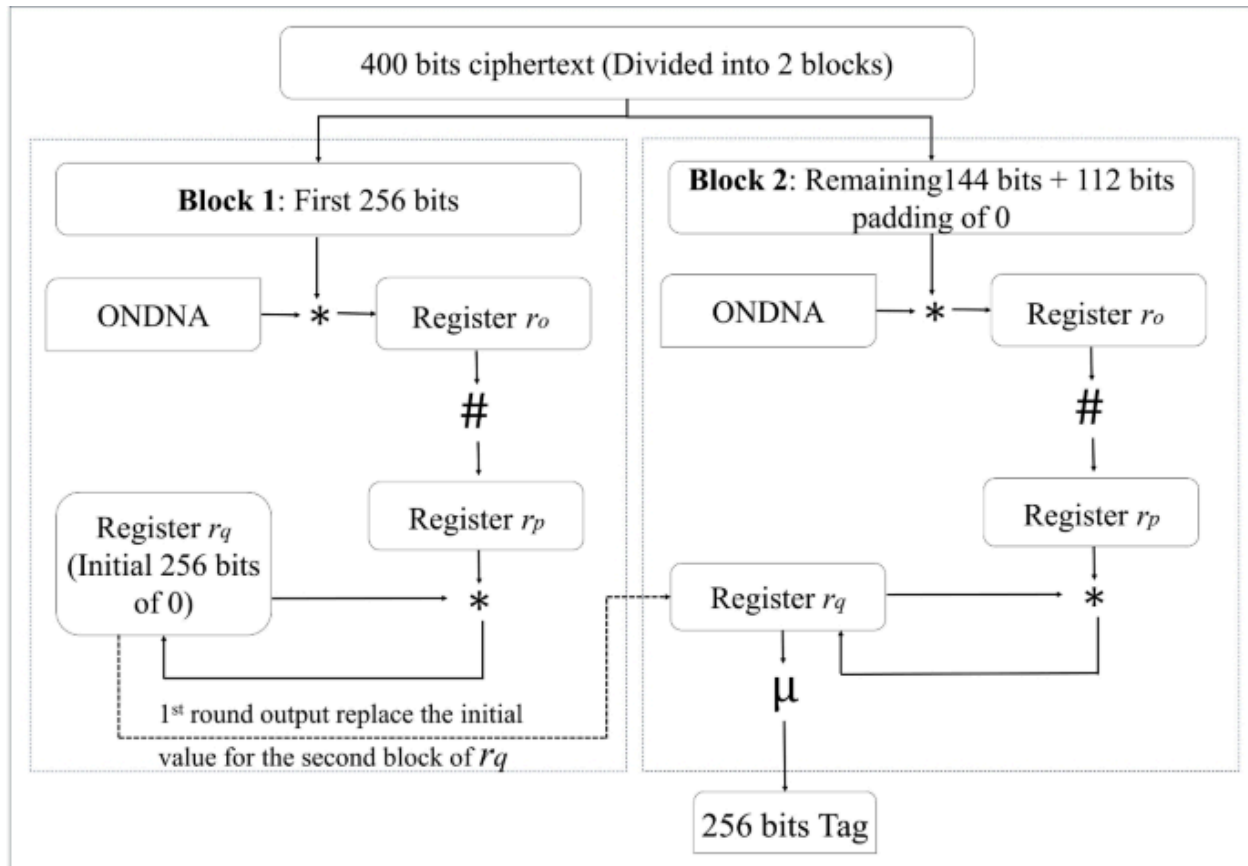
Assume a 400 bits message needs to be authenticated as shown in Fig. 5. The encryption process is described in Sect. "Omega network pseudorandom key generation based on DNA cryptography", and the output of encryption is the ciphertext (400 bits). Here, we further discuss the authentication process, which includes authentication key generation, tag generation, and tag verification. The authentication key generation process uses ONDNA, which produces a single block of 256 bits pseudorandom key-based DNA. Next, the 400 bits ciphertext is split into two blocks. The first block is 256 bits, while the second block is 144 bits. To align with the block size, the second block is padded with 112 bits of 0.

The first block of message is XORed with the 256 bits pseudorandom key-based DNA using DNA XOR rules (represented by $\oplus$), and the output is stored in register R_1 . Register R_1 is then processed using complementary mapping rules (represented by $\bar{\cdot}$), and the output is stored in register R_2 . Register R_2 is initialized with 256 bits of 0. Register R_2 is XORed with the initialized register using DNA XOR rules, and the result is stored in register R_3 .

The process is repeated for the second block. The second block is XORed with the same 256 bits symmetric secret key using DNA XOR rules, and the output is stored in register . Register is processed using complementary mapping rules, and the output is stored in register . Next, register is XORed with register using DNA XOR rules. The final result is processed using mRNA rules (represented by) to generate the 256 bits authentication tag.

Note that both the ciphertext and the tag are sent to receiver. In the verification algorithm, the receiver uses the ciphertext to generate the tag together with the same pseudorandom key-based DNA. The receiver then compares the generated tag with the tag t received from the sender. If the two tags are identical, such that , the ciphertext has not been altered by unauthorized parties. If the tag does not match, decryption becomes unnecessary since message integrity cannot be guaranteed. This approach minimizes the computational steps involved in decryption, optimizes the authentication process, and enhances efficiency by conserving computational resources.

Fig. 5



Symbol notes:

* DNA XOR rules

Complimentary mapping rules

μ mRNA rules

[Full size image](#)

Flow of the ODNAMAC process for 400 bits ciphertext demonstration.

ODNAMAC analysis and discussion

The code simulation for this research is developed using the programming language, chosen for its efficiency, flexibility, and widespread use in research simulation. The design of the ODNAMAC is simulated on a personal laptop equipped with an Intel i5-12500H processor running at 3.10 GHz, 12GB of RAM, and a 1TB SSD. The simulation aims to validate the effectiveness and security of the proposed ODNAMAC by testing its performance under various conditions, including different input sizes and collision attack scenarios. The results obtained from this simulation will provide valuable insights into the practical feasibility.

Tag generation

Two example test cases of the ODNAMAC are provided, along with their corresponding outputs. The inputs include a long message with minor changes, such that the first message lacks a full stop, while the second message includes a full stop. These results demonstrate that the proposed ODNAMAC generates distinct tags even when only a single character is changed in the input message.

- Message one: "Omega DNA Cryptography key-based authentication"
 - :
e3e300d3df5ee5e01b1bcc0b0f61161debebff4b434eb4e83f3f996
f6b63f638861dccbdb0 c15c12be52eee2e3a5
 - Ciphertext:
ac8e65b4be7ea1ae5a3b8f79761162728c999e293a6edf8d461ffb
0e18069218e7 68b8d5d5af287bdd339a8b8ccb
 - :
518ee59edde4777e206002c050039990a5523a033a2644e3e001
be7bbe1d663b
 - Tag:
db8afefc413cffaf498020fcc2c8528a1798f9afb15b36f04fab7966
cf97fb92

- Message two: "Omega DNA Cryptography key-based authentication."
 - :
 - 5557959d5959d5d52229c2c02d2c0252aaae8a0aada8aa3ae3
8e7ee0e8eebeb667b6b0 6b6b0666a889a8ac8a8ac8
 - Ciphertext:
 - 1a3af0fa3879919b630981b2545c763dcddceb68d488c15f97c3e
c1f938d8 a9ed712c2d80e05720fcbe8dcc5e5e4e6
 - :
 - 73ef3d3e3edfe366c61ae3e1e13a1ecd5434fcf3f3c43f0306b6929
b9b26b97b
 - Tag:
 - 2031a07d87f1b51a3f0d3bf8775de8a86bb694cdd0082b01ff9e31
187e5bb98d

The second simulation is the ODNAMAC is able to produce a fixed-length tag (256 bits) even when the input message is shorter than 256 bits.

- Message three: "Cryptography"
 - : 579985d7e9557e9d29cc6209
 - Ciphertext: 14ebfca79d3a19ef48bc0a70
 - :
 - af3ff3f633ee6f364515515211ee25125c8bcbcc18d7c719028f2f2
2580c 2c5a
 - Tag:
 - bb1cd9eeb2ac9b1784b6d8b044000d40dfbbfbff4b49f946f0bf0f0
0dbff 0fd3
- Message four: "Authentication"
 - : 527d9d2da5d77a777ec999e9b75d
 - Ciphertext: 1308e945c0b90e1e1da8ed80d833
 - :
 - 7f93f0dfffb309e3eca86ab3aaf6b81615f9449cff749934309a66a2
99c6a ab6b

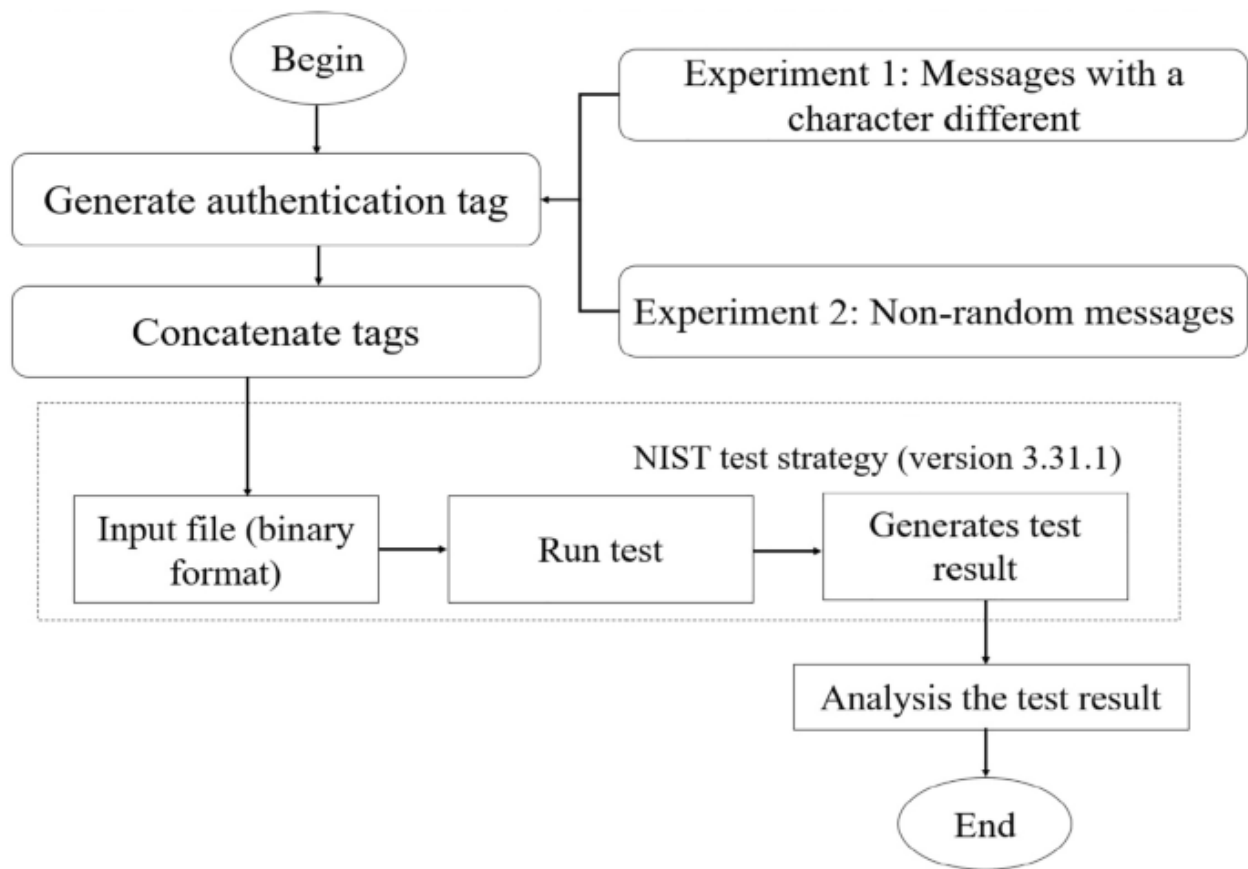
- Tag:
6da41fa0cf7fa85854157f40441afd6d8eaccaeeeccaaeeceea06605
aae6 00f6f

Dieharder statistical test

The Dieharder statistical testing suite is a collection of statistical tests designed to evaluate the quality and randomness of random number generators⁷. It is widely used in cryptography and simulations where high-quality randomness is critical¹⁸. In this research, the Dieharder statistical testing suite is used to test the tags generated by ODNAMAC. This ensures that even for similar messages, such as those differing by a single character, the tags produced will be different.

Figure 6 illustrates the process of the Dieharder statistical test. There are five steps: tags generation, tags concatenation, file selection, executing Dieharder, and obtaining the results. This process repeats twice for two experiments. In Experiment One, ten messages with each differing by one character is generated, ensuring slight variations in content. In Experiment Two, ten non-random messages are created, such as "000...000," "111...111," up to "999...999."

Fig. 6



[Full size image](#)

Dieharder statistical test process.

Next, each of the ten generated authentication tags is concatenated to form a binary file. Since each tag is 256 bits long, the total length of the concatenated tags is 2,560 bits. Then, one executes Dieharder version 3.31.1, selecting the binary file as input. To start the test, the command "dieharder -a -f filename.bin." is entered. Once the test completes, the results display, including the p-value and an assessment that indicates the randomness of the input data. The assessment is either PASSED or WEAK. A PASSED result indicates that the data is sufficiently random, while a WEAK result suggests that the input data may contain patterns or biases that attackers could exploit.

Fig. 7

```
File Actions Edit View Help
(kali@kali)-[~]
$ dieharder -a -f finaltag.bin
#
# dieharder version 3.31.1 Copyright 2003 Robert G. Brown
#
# rng_name | filename | rands/second |
# mt19937 | finaltag.bin | 4.89e+07 |
#
# test_name | ntupl | tsamples | psamples | p-value | Assessment
#
# diehard_birthdays | 0 | 100 | 100 | 0.23583000 | PASSED
# diehard_operm5 | 0 | 1000000 | 100 | 0.89088806 | PASSED
# diehard_rank_32x32 | 0 | 40000 | 100 | 0.12186044 | PASSED
# diehard_rank_6x8 | 0 | 100000 | 100 | 0.47689934 | PASSED
# diehard_bitstream | 0 | 2097152 | 100 | 0.46167356 | PASSED
# diehard_opso | 0 | 2097152 | 100 | 0.43622004 | PASSED
# diehard_oqso | 0 | 2097152 | 100 | 0.40545553 | PASSED
# diehard_dna | 0 | 2097152 | 100 | 0.61343105 | PASSED
# diehard_count_1s_str | 0 | 256000 | 100 | 0.92703637 | PASSED
# diehard_count_1s_byt | 0 | 256000 | 100 | 0.11394055 | PASSED
# diehard_parking_lot | 0 | 12000 | 100 | 0.54683180 | PASSED
# diehard_2dsphere | 2 | 8000 | 100 | 0.35447580 | PASSED
# diehard_3dsphere | 3 | 4000 | 100 | 0.59434530 | PASSED
# diehard_squeeze | 0 | 100000 | 100 | 0.76145860 | PASSED
# diehard_sums | 0 | 100 | 100 | 0.62049973 | PASSED
# diehard_runs | 0 | 100000 | 100 | 0.65503523 | PASSED
# diehard_runs | 0 | 100000 | 100 | 0.46122763 | PASSED
# diehard_craps | 0 | 200000 | 100 | 0.35928621 | PASSED
# diehard_craps | 0 | 200000 | 100 | 0.85564507 | PASSED
```

[Full size image](#)

Dieharder test result for concatenated 10 random messages' authentication tag.

Figure [7](#) displays the outcomes of the Dieharder statistical testing suite applied to 10 messages that differ by only one character, used as input to the proposed ODNAMAC scheme. The test produces a p-value of 0.23583, suggesting a relatively high probability that the generated outputs are uniformly random and free of detectable biases. This p-value indicates a low likelihood of collisions or predictable patterns in the outputs, thereby demonstrating the robustness and reliability of the ODNAMAC scheme in producing statistically sound and secure pseudorandom sequences.

Fig. 8

```
File Actions Edit View Help
(kali@kali)-[~]
$ dieharder -a -f nonrandom.bin

#=====#
# dieharder version 3.31.1 Copyright 2003 Robert G. Brown #
#=====#
# rng_name | filename | rands/second |
# mt19937 | nonrandom.bin | 4.28e+07 |
#=====#
# test_name | ntup | tsamples | psamples | p-value | Assessment |
#=====#
# diehard_birthdays | 0 | 100 | 100 | 0.11186905 | PASSED |
# diehard_operm5 | 0 | 1000000 | 100 | 0.85346349 | PASSED |
# diehard_rank_32x32 | 0 | 40000 | 100 | 0.44878600 | PASSED |
# diehard_rank_6x8 | 0 | 100000 | 100 | 0.87922180 | PASSED |
# diehard_bitstream | 0 | 2097152 | 100 | 0.97268506 | PASSED |
# diehard_opso | 0 | 2097152 | 100 | 0.17080731 | PASSED |
# diehard_oqso | 0 | 2097152 | 100 | 0.22403998 | PASSED |
# diehard_dna | 0 | 2097152 | 100 | 0.66075796 | PASSED |
# diehard_count_1s_str | 0 | 256000 | 100 | 0.37306148 | PASSED |
# diehard_count_1s_byt | 0 | 256000 | 100 | 0.85663413 | PASSED |
# diehard_parking_lot | 0 | 12000 | 100 | 0.06301159 | PASSED |
# diehard_2dsphere | 2 | 8000 | 100 | 0.51089560 | PASSED |
# diehard_3dsphere | 3 | 4000 | 100 | 0.94505307 | PASSED |
# diehard_squeeze | 0 | 100000 | 100 | 0.51996659 | PASSED |
# diehard_sums | 0 | 100 | 100 | 0.47971245 | PASSED |
# diehard_runs | 0 | 100000 | 100 | 0.48172129 | PASSED |
# diehard_runs | 0 | 100000 | 100 | 0.60812085 | PASSED |
# diehard_craps | 0 | 200000 | 100 | 0.64717625 | PASSED |
# diehard_craps | 0 | 200000 | 100 | 0.16197862 | PASSED |
```

[Full size image](#)

Dieharder test result for concatenated 10 non-random messages' authentication tag.

Similarly, Fig. [8](#) presents the Dieharder statistical testing suite results for 10 non-random messages used as input. The test yields a p-value of 0.11187. A p-value below 1 suggests that the probability of collisions is minimal, which is a critical factor for ensuring the security and reliability of the ODNAMAC scheme.

These results demonstrate that the proposed authentication mechanism is highly effective in generating unique authentication tags for distinct inputs. Specifically, the findings show that it is computationally difficult to find two different inputs that produce the same authentication tag. This property is essential for preventing collision attacks, where an attacker attempts to forge a valid tag for a different message. By ensuring that even small changes in the input message result in significantly different tags, the ODNAMAC scheme provides strong guarantees of message integrity and authenticity.

Formal security analysis

This section presents the formal security proof of the ODNAMAC scheme, demonstrating that the scheme (Gen, Tag, Ver) is secure. The proof establishes that ODNAMAC satisfies the required security properties, ensuring its resilience against existential forgery attacks.

Theorem 1

Suppose that ONDNA is an ideal pseudorandom function satisfying Definition [2](#). If ODNAMAC is constructed using ONDNA, then the ODNAMAC scheme (Gen, Tag, Ver) is secure against existential forgery under a chosen message attack with respect to the private key k with entropy and over the message space .

Proof

To meet the requirements stated in Theorem 1, we must demonstrate

1. the ONDNA is an ϵ -pseudorandom function.
2. the ODNAMAC (Gen, Tag, Ver) scheme is secure against existential forgery under a chosen message attack.

To prove (1), we assume that ONDNA is not a ϵ -pseudorandom function. This would mean that there exists a polynomial time adversary A capable of distinguishing a string of length n either it is a pseudorandom key-based DNA generated by ONDNA or a truly random string, all within polynomial time T . The probability of making the correct guess is greater than $\frac{1}{2} + \epsilon$, where ϵ is not negligible. This contradicts the assumption that ONDNA is a secure pseudorandom function generator satisfying Definition 2. Hence (1) is true. This implies that the private key k that is used in ODHAMAC can be distinguished with probability not greater than ϵ .

To prove (2) assuming the ODNAMAC(Gen, Tag, Ver) scheme is not secure against existential forgery under a chosen message attack. This means that there exists a polynomial time adversary A capable of producing a valid forgery with non-negligible probability. On the basis of adversary A we build the adversary B to break the pseudorandom security of ODNAMAC. Adversary B has access to an oracle O , which is either ODNAMAC($k, \cdot$) or a truly random function $f(\cdot)$. Adversary B simulates the ODNAMAC oracle for A using O . When adversary A queries a message m , adversary B computes t as the tag. If $t = \text{ODNAMAC}(k, m)$, this perfectly simulates the ODNAMAC scheme. Eventually, adversary A outputs a forgery m^* , where m^* is not being queried before. Adversary B checks if $t^* = \text{ODNAMAC}(k, m^*)$. If so, adversary B outputs 1, guessing $k = \text{ODNAMAC}(k, \cdot)$; otherwise, adversary B outputs 0, guessing f . Note that if $k = \text{ODNAMAC}(k, \cdot)$, adversary B succeeds in

forging with non-negligible probability, so adversary outputs 1 with probability ϵ . If $\mathcal{A}$ succeeds with probability at most ϵ . Thus, adversary distinguishes $\text{ODNAMAC}(k, \cdot)$ from $f(\cdot)$ with non-negligible advantage ϵ . This contradicts the assumption that ONDNA is a secure pseudorandom function. Hence, by Theorem 1 the $\text{ODNAMAC}(\text{Gen}, \text{Tag}, \text{Ver})$ scheme built from the ONDNA must be secure against existential forgery under a chosen message attack, satisfying Definition 3. Hence (2) is true.

Hence, by Theorem 1 the ODNAMAC that is built from the ONDNA is $(\text{Gen}, \text{Tag}, \text{Ver})$ is secure against existential forgery under a chosen message attack with respect to the private key k with entropy ϵ and over the message space $\mathcal{M}$.

Brute force attack

A brute force attack is a straightforward search method that seeks to identify the correct input by systematically trying every possible option within the input space. The input can be message or private key. This approach is typically used when the attacker cannot exploit specific vulnerabilities or weaknesses in the cryptographic scheme. The attacker must carefully test all potential input values until the correct one is found.

Since the proposed ODNAMAC is designed to satisfy Definitions 2 and 3, an adversary who has the tag generated by ODNAMAC could attempt a brute force attack on either the message or the private authentication key. Assume a message m has length of $|m|$. If the length of the message m is shorter than the private authentication key, then discovering the correct message generally requires fewer attempts than uncovering the private authentication key. If the message length exceeds 256 bits, the security complexity of ODNAMAC is at most 2^{256} . Conversely, if the message length is less than 256 bits, the security complexity is $2^{|m|}$.

Collision attack

Assume a message m has length of $m/$ and a tag generation process tag maps m to an output a tag t of length 256 bits. Collision will occur when $m/$ exceeds 256 bits. For the tag generation process tag on a random message m , a message collision occurs when . When the output length is 256 bits, the birthday paradox²² indicates that after calculating tag for approximately distinct messages, there is a 50% chance of a collision. Therefore, one may conclude that the security complexity of ODNAMAC against collision attacks is approximately .

Conclusion

DNA cryptography is widely recognized as a lightweight cryptographic method characterized by its low computational complexity, making it particularly suitable for resource constrained environments such as IoT devices, embedded systems, and mobile applications. This research introduces the Omega DNA cryptography key-based authentication scheme (ODNAMAC), a novel and innovative approach firmly grounded in the principles of DNA cryptography. This mechanism not only preserves the confidentiality of data but also introduces robust message authentication capabilities. The proposed mechanism employs the ONDNA approach to produce a fixed length authentication tag of 256 bits from arbitrary length of input. It passed the Dieharder statistical test. Overall, this paper establishes that the ODNAMAC is proven (Gen, Tag, Ver) secure against existential forgery under a chosen message attack with respect to the private key k with entropy and over the message space . The current work is only able to identify unauthorized modifications without privacy preservation. For future work, one may extend ODNAMAC to include privacy preserving authentication.

Data availability

Data authentication generated during this study are included in this published article. Security analysis datasets used and/or analysed for current study available from the corresponding author on reasonable request.

Code availability

Share upon request.

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Bio-inspired cryptography based on proteinoid assemblies

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Editor: Daniel Ioan Hunyadi³

PMCID: PMC12118996 PMID: [40435278](#)

Abstract

We present an innovative cryptographic technique inspired by the self-assembly processes of proteinoids—thermally stable proteins that form spontaneously under prebiotic conditions. By emulating the deterministic yet complex interactions within proteinoid assemblies, the proposed method generates secure encryption keys and algorithms. We measure the unique electrical properties of proteinoid microspheres. Their capacitance values range from -656.6 to 434.9 nF. Then, we convert these measurements into encryption keys using the formula $k_i = (\lfloor |C_i| \cdot 100 \rfloor \bmod 256)$. The approach harnesses the inherent unpredictability of proteinoid behavior to create a robust and adaptable encryption framework resilient to cryptanalytic attacks. The encryption process uses modular multiplication: $e_j = (m_j \cdot k_i) \bmod 256$. This changes plaintext into ciphertext. The security relies on electrical signatures that depend on the composition. Experimental results show that this bio-inspired system aligns with contemporary encryption standards, offering significant benefits in key generation and distribution. Our implementation has a linear computational complexity of $O(n)$. It offers security levels ranging from 8 to 128 bits, based on composition. Additionally, it is energy efficient, performing about 200 operations per joule. Statistical analysis further affirms the high randomness of the generated keys, highlighting the potential of biological processes in advancing cryptographic security.

1 Introduction

Cryptography, a cornerstone of secure communication, has been integral to human society for centuries. As technology continues to evolve and the demand for robust security measures increases, researchers are exploring innovative cryptographic techniques inspired by a range of disciplines, including biology. Bio-inspired cryptography has emerged as a prominent field of study, leveraging the intricate and unique properties of biological systems to develop advanced data protection methods [1–10]. This field can be broadly classified into several key areas [11]. DNA Cryptography uses DNA's structure for cryptographic tasks [12–15]. These include data storage, encryption, and key generation. DNA's high density and parallelism give cryptographic systems unique benefits. Behavioral Biometrics uses unique biological patterns, such as typing rhythms and EEG signals, for security. Their uniqueness makes them well-suited for biometric authentication [16–19]. Swarm Intelligence Cryptography [20–22] draws inspiration from social insects, such as ants, for cryptographic tasks, including key distribution and secure communication. The decentralized, self-organizing nature of swarms may enable strong and resilient systems. Finally, Bio-inspired algorithms [23–25] are cryptographic algorithms derived from biological processes, such as evolution and neural networks. They are often adaptable and robust, making them well-suited for complex cryptographic applications. This emerging field uses biological principles that have evolved over millions of years to solve complex security problems [26,27]. Biological systems have strong data protection methods [12]. They can repair DNA defects, detect immune systems, and learn from evolutionary networks [28]. This approach lays a strong foundation for new algorithms. They can be more secure, efficient, and scalable than traditional methods. This field is important. It can solve challenges using biologically proven solutions [29]. This is especially true for key generation, authentication, and secure communication protocols [30–32]. The use of proteinoid assemblies represents an interesting area of exploration in bio-inspired cryptography. Proteinoids are synthetic polymers designed to replicate the structure and function of natural proteins [33]. The assemblies demonstrate notable characteristics, including self-organization, molecular recognition, and information processing, making them suitable for cryptographic applications [34]. Recent advancements in bio-inspired cryptography have been important, as researchers investigate the potential applications of various biological systems for secure communication. DNA-based cryptography has attracted significant attention owing to the high information density and parallel processing capabilities built into DNA molecules [35]. Furthermore, the application of cellular automata, which are

discrete models derived from biological cells, has been explored for cryptographic applications [[36](#)].

Proteinoid assemblies have not been extensively examined within the field of cryptography; however, their distinctive characteristics and similarities to natural proteins indicate significant potential. Proteinoids exhibit the ability to self-assemble into stable configurations, including microspheres and vesicles, which serve to encapsulate and protect information [[37](#)]. Additionally, the capacity of proteinoids to identify and attach to particular molecules can be used for targeted encryption and decryption operations [[38](#)].

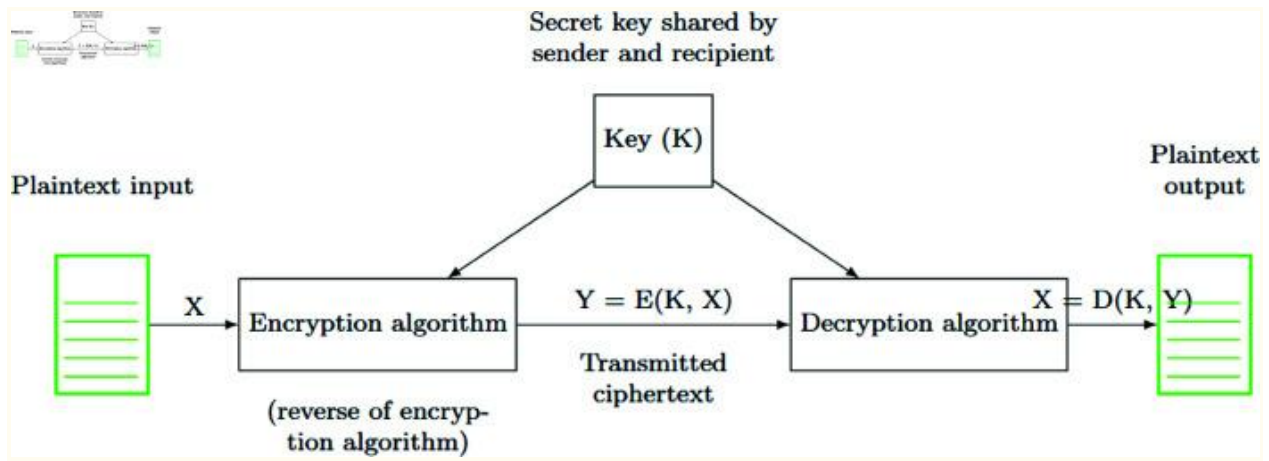
The complexity and diversity of proteinoid assemblies offer a solid basis for the development of secure cryptographic systems. The extensive variety of potential proteinoid sequences, along with the sensitivity of their assembly to environmental conditions, provides a significant key space for encryption [[39](#)]. Additionally, the dynamic characteristics of proteinoid assemblies, including their capacity for conformational changes and responsiveness to external stimuli, can be used for adaptive and context-aware cryptography [[40](#)].

Bio-inspired cryptography utilising proteinoid assemblies presents a potential solution to contemporary cryptographic challenges, particularly the demand for lightweight and energy-efficient algorithms that are appropriate for resource-constrained devices [[41](#)]. The properties of self-assembly and molecular recognition in proteinoids may facilitate the creation of cryptographic systems that are both self-organising and self-authenticating [[42](#)].

The study of proteinoid assemblies for cryptography corresponds with the overarching trend of incorporating biological principles into computing and information processing. Researchers aim to develop innovative and sustainable technologies by drawing inspiration from the sophistication and efficiency of natural systems [43]. The intersection of biology and cryptography presents significant potential for the development of secure communication systems characterized by robustness, adaptability, and scalability. This paper presents a novel bio-inspired cryptographic framework that uses proteinoid assemblies. This study examines the distinct characteristics of proteinoids and their applicability in encryption and decryption methodologies. We demonstrate the feasibility and security of proteinoid-based cryptography through experimental studies and theoretical analysis. The objective of this work is to advance the field of bio-inspired cryptography and facilitate subsequent research and development efforts in this area.

This research investigates an innovative method of symmetric cryptography utilising proteinoid-based systems. [Fig 1](#) depicts the essential principle of symmetric encryption and decryption. This procedure uses a shared secret key, K , for both encryption, $E(K,X)$, and decryption, $D(K,Y)$. It is the basis of our proteinoid-inspired cryptographic system. Our approach improves this model. It extracts the key K from the unique electrical traits of proteinoid compositions, especially their capacitance. This bio-inspired method may improve key creation and management. The following sections will explore this further.

Fig 1. Symmetric encryption and decryption procedure.



[Open in a new tab](#)

The plaintext input X undergoes encryption via the method E with a secret key K , resulting in ciphertext $Y = E(K, X)$. The system delivers the ciphertext and then decrypts it with method D using the same key K . This recovers the original plaintext $X = D(K, Y)$. The secret key K is safely disclosed between the sender and recipient before communication occurs. This procedure ensures confidentiality. Only those who know K can decrypt the communication. The same key is used for both encryption and decryption. This shows the system's symmetry.

[Table 1](#) illustrates a comparison of various bio-inspired encryption technologies, detailing their features, advantages, and limitations. [Table 1](#) compares biologically inspired cryptography techniques. Each has its own benefits but supports traditional cryptography. Traditional cryptography uses math and established protocols [44]. Biologically inspired methods use new

techniques from biology to boost security. DNA-based cryptography uses the vast power of DNA’s parallel processing. It also uses traditional key functions. Cellular automata-based methods create complex, unpredictable patterns. These can enhance existing cryptographic primitives. This complementary relationship lets systems use both methods. They get the core security of traditional approaches and the new abilities of bionic ones. This results in deep cryptographic hybrid solutions.

Table 1. Comparison of biological encryption technologies. This table summarizes various bio-inspired encryption methods, highlighting their features, advantages, and limitations. DNA-based cryptography uses DNA’s high information density and parallel processing for secure communication. Cellular automata encryption employs the dynamic properties of these systems, inspired by biological cells, to create complex patterns. Chaotic systems, mimicking unpredictable biological processes, are used in cryptography due to their sensitivity to initial conditions. Evolutionary algorithms, inspired by biological evolution, aim to enhance cryptographic methods. However, these methods face challenges like scalability, implementation issues, and the need for more research to ensure security. The proposed proteinoid-based cryptography in this paper aims to utilize the unique properties of proteinoid assemblies—self-organization and molecular recognition—to create a new, secure encryption system.

Technology	Key Features	Advantages and Limitations
------------	--------------	----------------------------

DNA-based cryptography [35]	- Exploits high information density of DNA - Parallel processing capabilities - Molecular-level encryption - Computational complexity: $O(n^2)$ - Security level: 70-128 bits - Energy efficiency: ~ 10 ops/J	- High storage capacity - Potential for DNA computing - Challenges in practical implementation - Susceptible to errors and mutations
Cellular automata-based encryption [45]	- Discrete models inspired by biological cells - Dynamic behaviour and complex patterns - Local interactions and global emergence - Computational complexity: $O(n)$ - Security level: 80-120 bits - Energy efficiency: ~ 1000 ops/J	- Parallel and distributed processing - Generates complex encryption patterns - Scalability and computational efficiency - Limited research on cryptanalytic resistance
Chaotic systems [45]	- Exhibits unpredictable and random-like behaviour - Sensitive to initial conditions - Inspired by chaotic processes in biology - Computational complexity: $O(n \log n)$ - Security level: 60-100 bits - Energy efficiency: ~ 500 ops/J	- High sensitivity to initial conditions - Generates random-like sequences - Potential for secure communication - Requires careful parameter selection
Evolutionary algorithms [46]	- Inspired by biological evolution - Optimization and adaptation mechanisms - Population-based search and selection - Computational complexity: $O(n^2 \log n)$ - Security level: Variable (40-90 bits) - Energy efficiency: ~ 5 ops/J	- Optimizes cryptographic primitives - Adapts to changing security requirements - Computationally intensive - Convergence to global optimum not guaranteed
Proteinoid-based cryptography (proposed)	- Self-organization and molecular recognition - Vast key space and dynamic behaviour - Adaptive and context-aware encryption - Computational complexity: $O(n)$ - Security level: 8-128 bits	- Unique properties of proteinoid assemblies - Potential for lightweight and energy-efficient algorithms - Self-organizing and self-authenticating systems - Experimental

(composition dependent) - Energy efficiency: ~ 200 ops/J validation and further research required

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The computational complexity notation (e.g., $O(n^2)$) describes how an algorithm's resource requirements scale with the input size n . When the input size doubles, the computational needs grow approximately four times. This scaling behavior often shows up in algorithms with nested loops. In these cases, the whole dataset is processed several times. The security level, measured in bits (e.g., 70--128 bits), quantifies cryptographic strength. To break a system with 128-bit security, an attacker would need to perform roughly 2^{128} operations. This number is so enormous that current technology makes achieving it almost impossible. A lower bound of 70 bits means weaker protection. This can make the system vulnerable to advanced attacks. Security can vary based on specific implementation details or methods. Energy efficiency (e.g., ~ 10) ops/J (operations per joule). This metric shows how much computational work we perform for each unit of energy used. In resource-limited settings like mobile apps, power use is key. It has a significant effect on the app's performance. Higher values show better energy efficiency. Modern cryptographic systems strive to maximize this ratio while maintaining robust security levels.

Cellular automata-based encryption and our proteinoid-based method both scale linearly ($O(n)$). This makes them the most efficient choices in terms of computation. This linear relationship means that processing time increases with input size. So, performance stays predictable as data volumes grow.

Chaotic systems have a complexity of $O(n \log n)$. In contrast, DNA-based cryptography runs at $O(n^2)$, and evolutionary algorithms are even more demanding at $O(n^2 \log n)$. These higher complexities can limit their use in large datasets or resource-limited settings.

Security levels vary considerably across these bio-inspired methods. DNA-based cryptography (70–128 bits) and cellular automata (80–120 bits) offer security similar to well-known cryptographic protocols. Chaotic systems provide moderate security, ranging from 60 to 100 bits. In contrast, evolutionary algorithms offer the lowest guaranteed security, with a range of 40 to 90 bits. Our proteinoid-based approach shows the widest security range (8–128 bits), reflecting its composition-dependent nature. The lower bound shows single-sample setups. The upper bound shows composite setups. These use multiple proteinoid samples to reach security levels that meet modern standards.

Energy efficiency demonstrates perhaps the most striking differences. Cellular automata-based encryption is the most efficient, with around 1000 operations per joule. Next, we have chaotic systems that use about 500 ops/J. Our proteinoid-based method follows at around 200 ops/J. DNA-based cryptography uses roughly 10 ops/J, and evolutionary algorithms need about 5 ops/J. This difference is due to their complex biochemical and iterative computing needs. This parameter matters a lot for IoT devices, embedded systems, and other energy-limited settings.

Our proteinoid-based cryptography shows a strong balance. It has linear computational complexity, adjustable security levels, and moderate energy efficiency. Cellular automata are more energy-efficient now. However, our method may offer better adaptability and quantum resistance because it's

based on biology. Plus, there's still room to improve as the technology develops.

Our cryptographic framework works in a structured way. It uses the electrical properties of proteinoid assemblies. We start by making proteinoid microspheres. This happens through thermal polymerization of amino acids at 180 ± 1 °C. Next, we check their electrical properties. We measure resistance (R), impedance (Z), and capacitance (C) with precision LCR meters at 300 kHz. The capacitance values range from -656.6 to 434.9 nF. These values depend on the amino acid composition. They are used to generate cryptographic keys with the formula $k_i = (\lfloor |C_i| \rfloor \cdot 100) \bmod 256$. Encryption occurs via the function $e_j = (m_j \cdot k_i) \bmod 256$, where m_j represents plaintext ASCII values. This procedure creates an encryption system that is both predictable and unpredictable. Its security comes from the complex interactions of molecules in the proteinoid structures. We use several evaluation metrics: computational complexity ($O(n)$), security level (8-128 bits based on composition), and energy efficiency (about 200 operations per joule). This makes our method a biologically-inspired option compared to regular symmetric encryption algorithms.

2 Methods and materials

We made proteinoids using methods from our earlier study [47]. First, we heated amino acids to start polymerization, creating polymer chains. Then, these proteinoids formed microspheres in water at moderate temperatures. We generated input voltage bytes with a 10MHz Dual Channel Function/Arbitrary Waveform Generator (Model BK4053B). The input signal $V(t)$ consists of square pulses.

$$V(t)=\tag{1}$$

$$N-1$$

$$\Sigma$$

$$n=0$$

$$A\cdot\mathrm{rect}($$

$$t-nT$$

$$\tau$$

)

Here, A is the pulse amplitude, N is the number of pulses, T is the time between pulses, τ is the pulse width, and $\text{rect}(x)$ is the rectangular function. The signal was sampled at 20 MS/s, giving a time resolution of $\Delta t = 50$ ns. We varied τ and T to create a pseudo-random sequence. Over 52.43 ms, we collected 1,048,573 samples. This high-frequency, pseudo-random signal was key to studying the electrical properties of proteinoid microspheres. We used a BK Precision LCR meter (Model 891) at 300 kHz to characterize the proteinoids electrically. This device accurately measured the capacitance (C), resistance (R), and impedance (Z) of the samples. The complex impedance Z of the proteinoid samples is expressed as:

$$Z = R + jX_C = R - j \tag{2}$$

1

$$2\pi fC$$

Here, f is the frequency (300 kHz), R is the resistance, and X_C is the capacitive reactance. These measurements gave us key insights into the electrical properties of proteinoid microspheres. This assessment helps determine their potential in bio-inspired electronic devices.

To generate Power Spectral Density (PSD) plots, we employed MATLAB's *pwelch* function, which estimates the PSD of the input signal using Welch's method. The PSD, denoted as $P_{xx}(f)$, is calculated as:

$$P_{xx}(f) = \frac{1}{N} \sum_{k=1}^N |X_k(f)|^2 \quad (3)$$

1

N

$$N$$

$$\Sigma$$

$$i=1$$

$$|$$

$$L-1$$

$$\Sigma$$

$$n=0$$

$$w(n)x_i(n)e^{-j2\pi fn} \big|_2$$

where N is the number of segments, L is the segment length, $w(n)$ is the window function, and $x_i(n)$ is the i -th segment of the input signal. The resulting PSD is plotted on a logarithmic scale:

$$\text{PSD (dB/Hz)} = 10 \log_{10}(P_{xx}(f)) \quad (4)$$

The frequency axis is normalized to MHz for clarity:

$$f_{\text{MHz}} = \quad (5)$$

$$f$$

To quantify the spectral characteristics, we calculated key metrics such as the peak frequency (f_{peak}), peak power (P_{peak}), and total power (P_{total}):

$$f_{\text{peak}} = \text{argmax}_f P_{xx}(f) \quad (6)$$

$$P_{\text{peak}} = 10 \log_{10}(\max P_{xx}(f)) \quad (7)$$

$$P_{\text{total}} = 10 \log_{10}(\quad) \quad (8)$$

Σ f $P_{xx}(f)$

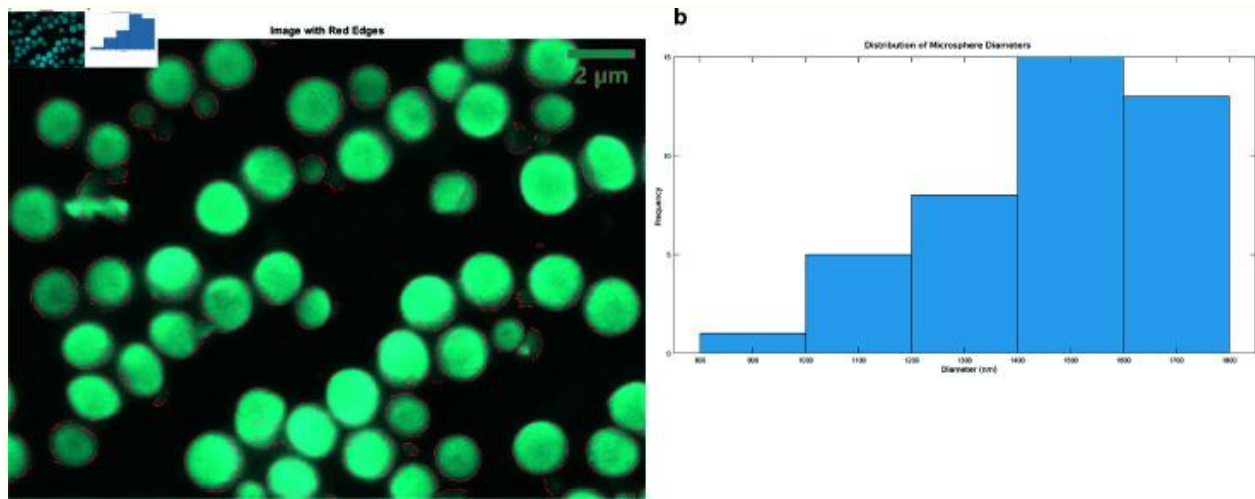
The metrics provide a way to measure the spectral variations in proteinoid assemblies. They may help in generating cryptographic keys. Each proteinoid type has a unique mix of peak frequencies, power distributions, and total spectral energy. This makes our bio-inspired encryption technique more reliable.

3 Results

3.1 Morphological characteristics of proteinoid assemblies

[Fig 2](#) shows the features and sizes of proteinoid microspheres. The SEM image ([Fig 2a](#)) reveals uniform spherical shapes. Edge detection identifies microsphere boundaries, allowing size measurements. The size ranges from 800 nm to 1800 nm, averaging 1457.03 ± 219.46 nm ([Fig 2b](#)). It has a slight negative skewness (-0.4844) and a kurtosis of 2.1773 . This suggests the microspheres have a flatter and larger shape compared to the average. These stats reflect the synthesis process's consistency and control. The narrow size range and uniform shape imply a careful synthesis method. This is vital for drug delivery and specialized materials.

Fig 2. Analysis of the morphology and statistics of proteinoid microspheres.

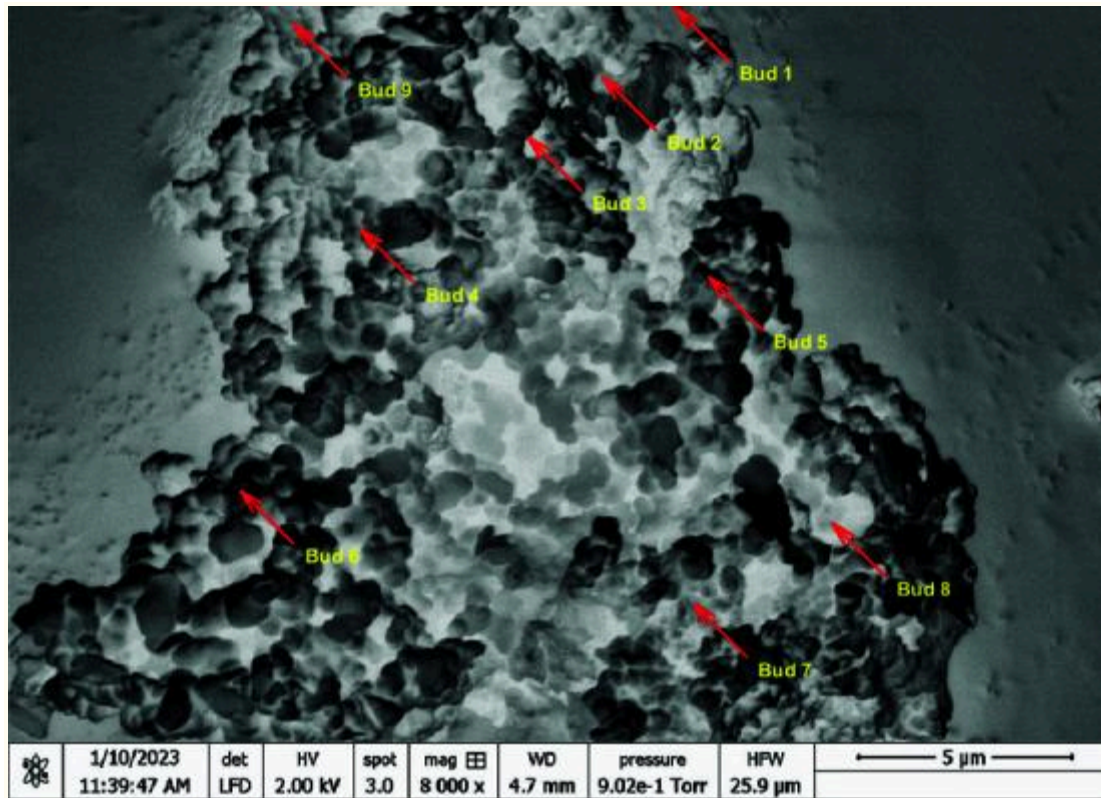


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(**a**) SEM image with edge detection emphasising the spherical morphology of the microspheres. The scale bar denotes $2\text{ }\mu\text{m}$. (**b**) Histogram illustrating the distribution of microsphere diameters. The distribution spans roughly 800 nm to 1800 nm, with a mean diameter of $1457.03 \pm 219.46\text{ nm}$. The distribution has a slight negative skewness (-0.4844) and is platykurtic (kurtosis 2.1773). It is somewhat flat-topped and tends to have wider diameters.

The budding process observed in proteinoid microspheres, as illustrated in [Fig 3](#), can be characterized by the equation:

Fig 3. High-resolution scanning electron microscopy (SEM) image of proteinoid microspheres exhibiting budding reproduction.



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The image, taken at 8000× magnification (HV = 2.00 kV, spot size = 3.0), shows the complex surface and budding of proteinoid structures. Annotators have labeled nine distinct buds (Bud 1 to Bud 9). They highlight the asexual reproduction of these biomimetic entities.

$$P_{\text{parent}} \rightarrow P_{\text{parent}} + P_{\text{bud}} \quad (9)$$

where P_{parent} represents the parent proteinoid and P_{bud} the newly formed bud. This process demonstrates the self-replicating capability of these structures. We took the SEM image at high vacuum (9.02×10^{-1} Torr) with a 4.7 mm working distance. This ensured optimal resolution for observing nanoscale features. The $5 \mu\text{m}$ scale bar shows that the proteinoid aggregate is about 15-20 μm long. The Harmony Finder Width (HFW) of $25.9 \mu\text{m}$ confirms the field of view. This new phenomenon, like cell division, suggests a way for proteinoid microspheres to grow and spread. In bio-inspired cryptographic systems, attackers can exploit this process. The general encryption function describes it:

$$E(m, k_{\text{proteinoid}}) = c \quad (10)$$

where m is the message, $k_{\text{proteinoid}}$ is the key derived from proteinoid properties, and c is the resulting ciphertext. The unique electrical and cryptographic properties of proteinoid assemblies come from their diverse shapes and sites for budding. Researchers observe budding as a reproductive method in both proteinoid microspheres and various organisms. In yeast, for example, the equation describes the budding process:

$$Y_{\text{parent}} \rightarrow Y_{\text{parent}} + Y_{\text{daughter}} \quad (11)$$

where Y_{parent} is the parent yeast cell and Y_{daughter} is the newly formed daughter cell [48]. This process involves asymmetric cell division. A smaller daughter cell emerges from a larger parent cell. Similarly, in some cnidarians, like Hydra, asexual reproduction through budding is:

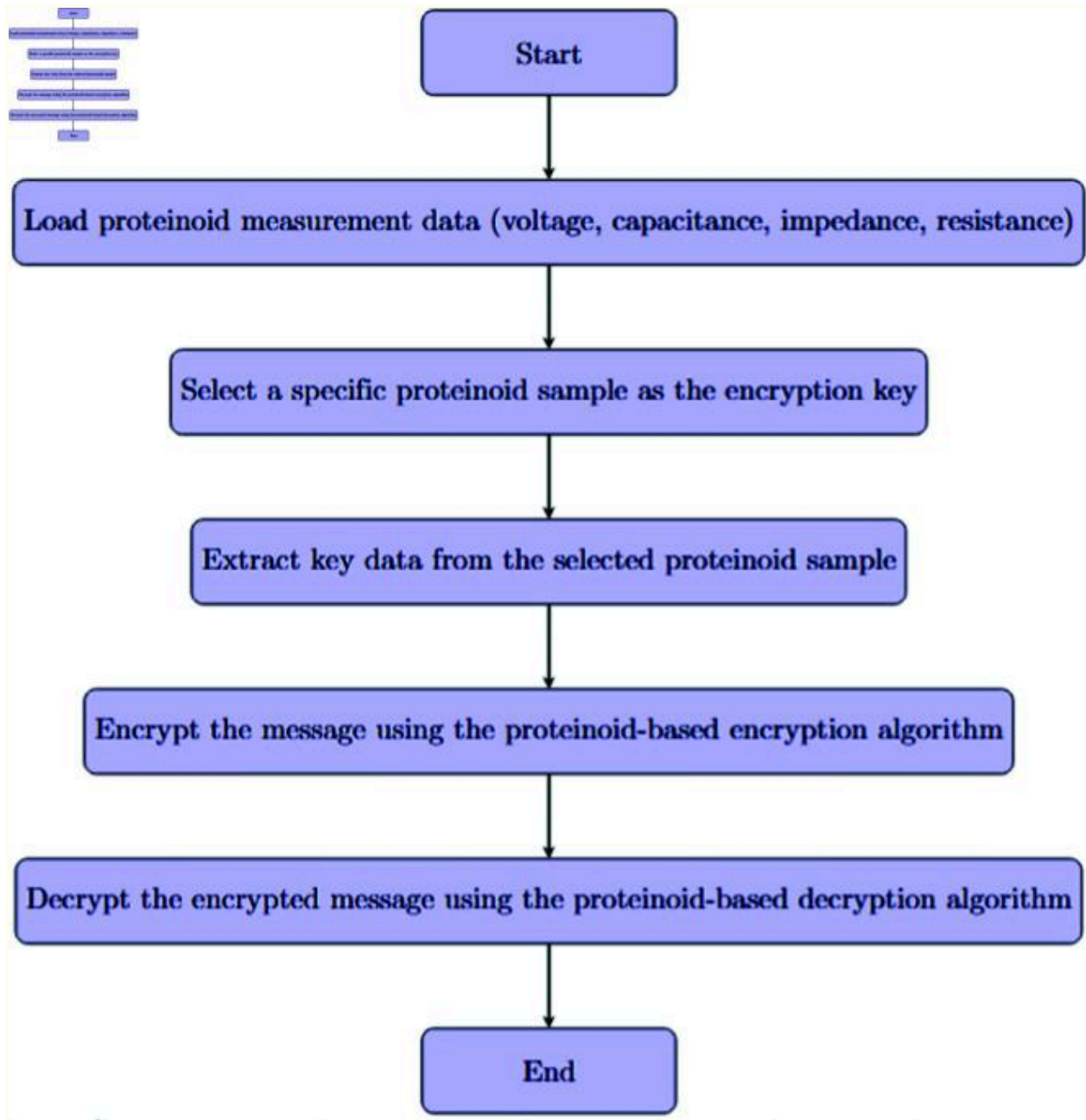
$$H_{\text{parent}} \rightarrow H_{\text{parent}} + H_{\text{bud}} \quad (12)$$

here H_{parent} is the parent Hydra and H_{bud} is the newly formed bud [49]. In Hydra, budding occurs through the formation of a small outgrowth on the body column, which develops into a miniature individual before detaching. The budding in proteinoid microspheres (Equation 9) is like biological processes (Equations 11 and 12). This similarity highlights proteinoids' biomimetic properties. It also opens up potential applications in drug delivery and artificial cell development, beyond cryptography.

3.2 Proteinoid-based encryption: a novel bio-inspired approach

Our proteinoid-based encryption method marks a big advance in bioinspired cryptography. Figure 4 shows how this new method uses proteinoids' unique electrical properties. It creates a strong, adaptable encryption scheme. The procedure outlined in the generic proteinoid encryption method flowchart (Fig. 4) illustrates the utilisation of biological molecules to secure communication.

Fig 4. Proteinoid-based encryption process. The setup involves importing measurement data, including voltage, capacitance, impedance, and resistance, from various proteinoid samples.



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A designated proteinoid sample is selected as the encryption key based on its unique electrical properties. Selection looks at Shannon entropy for derived keys, aiming for $H(K) > 7.5$ bits/byte. It also checks measurement reproducibility, needing over 98% consistency in repeated tests. Lastly, it assesses temporal stability, with less than 1% drift over 30 days. Proteinoid compositions with capacitance values over 100 nF offer the best security features. They serve as encryption keys thanks to their unique electrical properties. The encryption algorithm utilizes these characteristics to secure the original message, while the decryption algorithm applies the same key to retrieve the original data. This method leverages the consistent and distinct electrical signatures of proteinoids to enable a robust encryption scheme.

The initial phase of our method has a key task. It is to load proteinoid measurement data, which includes voltage, capacitance, impedance, and resistance (see [Fig 4](#)). This careful data collection will capture all the electrical traits that distinguish each proteinoid sample. The mix of electrical properties makes encryption keys more complex and secure. This helps resist attacks.

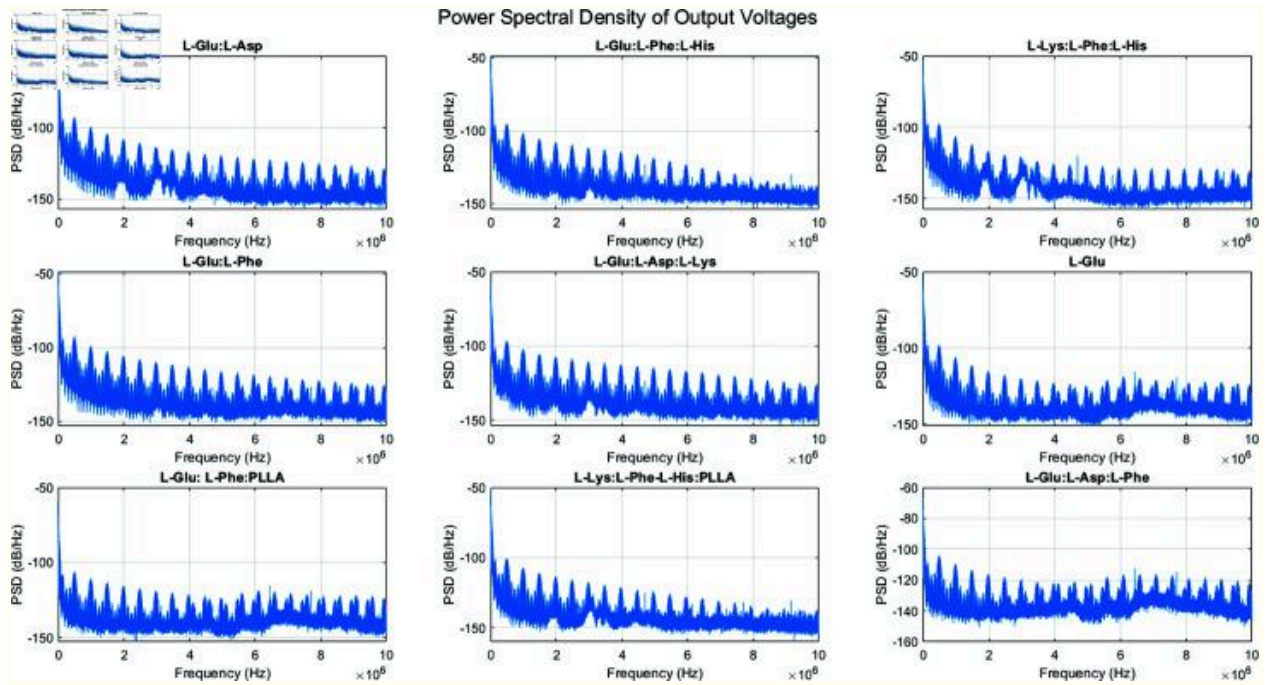
The core of the encryption process is centred on the selection and extraction of key data from a designated proteinoid sample ([Fig 4](#)). This step is crucial. It turns the proteinoid's biology into a usable encryption key. [Fig 4](#) presents encryption and decryption algorithms that effectively utilize the keys derived from proteinoids. This method secures the encrypted message. It adds a new, biological element to cryptography.

[Fig 4](#) shows the cyclical process of encryption and decryption. It highlights their use in secure communication. Using various proteinoid samples as keys,

as shown in the flowchart, improves our method's security and flexibility. This feature, and the unique electrical behaviour of proteinoids, create a new encryption tech. It integrates biological principles with cryptographic methods.

We analyzed the electrical and spectral spikes of various proteinoid assemblies. This was to assess their potential in bio-inspired cryptography. [Fig 5](#) shows the PSD of the output voltages for nine different proteinoid samples:

Fig 5. Power spectral density (PSD) of output voltages for various proteinoid samples.



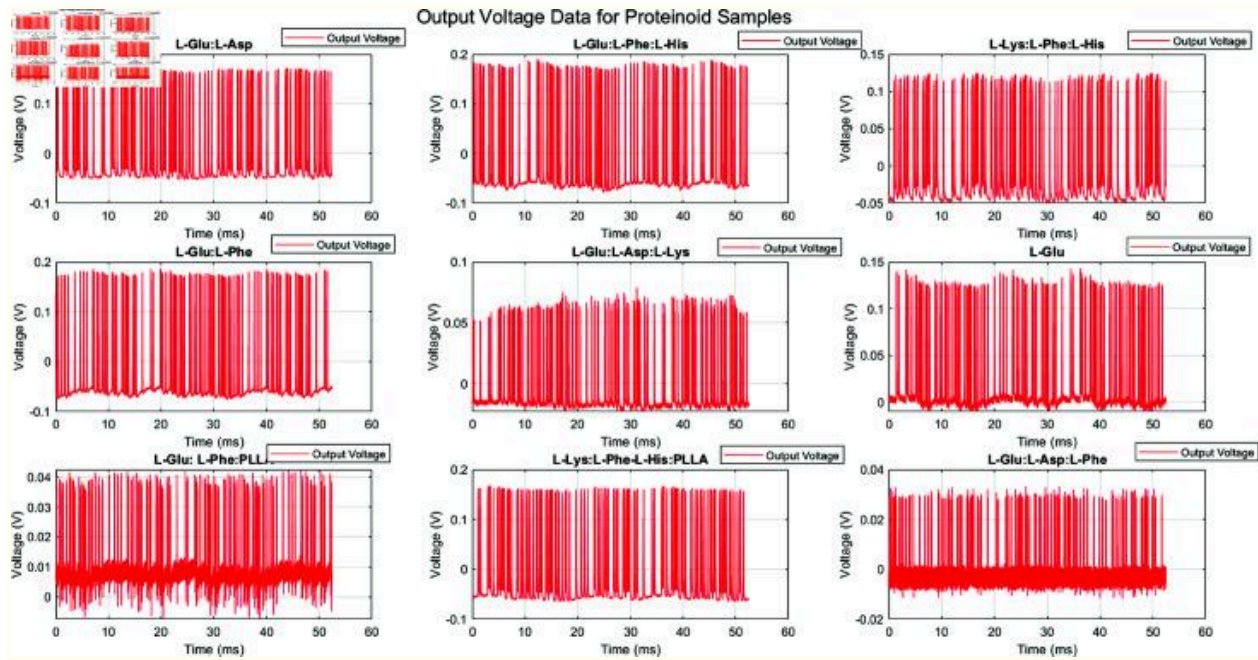
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The figure shows the frequency-domain analysis of the output voltage signals from nine different proteinoid compositions. Each subplot displays the PSD, measured in dB/Hz, across a frequency range of 0 to 10 MHz. The analyzed proteinoid samples include: L-Glu:L-Asp, L-Glu:L-Phe:L-His, L-Lys:L-Phe:L-His, L-Glu:L-Phe, L-Glu:L-Asp:L-Lys, L-Glu, L-Glu:L-Phe:PLLA, L-Lys:L-Phe-L-His:PLLA, and L-Glu:L-Asp:L-Phe.

The PSD plots reveal distinct spectral signatures for each proteinoid assembly. All samples show a general trend of decreasing power with increasing frequency, but with unique patterns of peaks and troughs. For example, L-Glu:L-Asp:L-Phe has more high-frequency components than the others. L-Glu shows a smoother decay across the spectrum. These unique spectral fingerprints could serve as a valuable resource for cryptographic applications.

[Fig 6](#) displays the time-domain output voltage signals for the same proteinoid samples:

Fig 6. Output voltage data for various proteinoid samples.



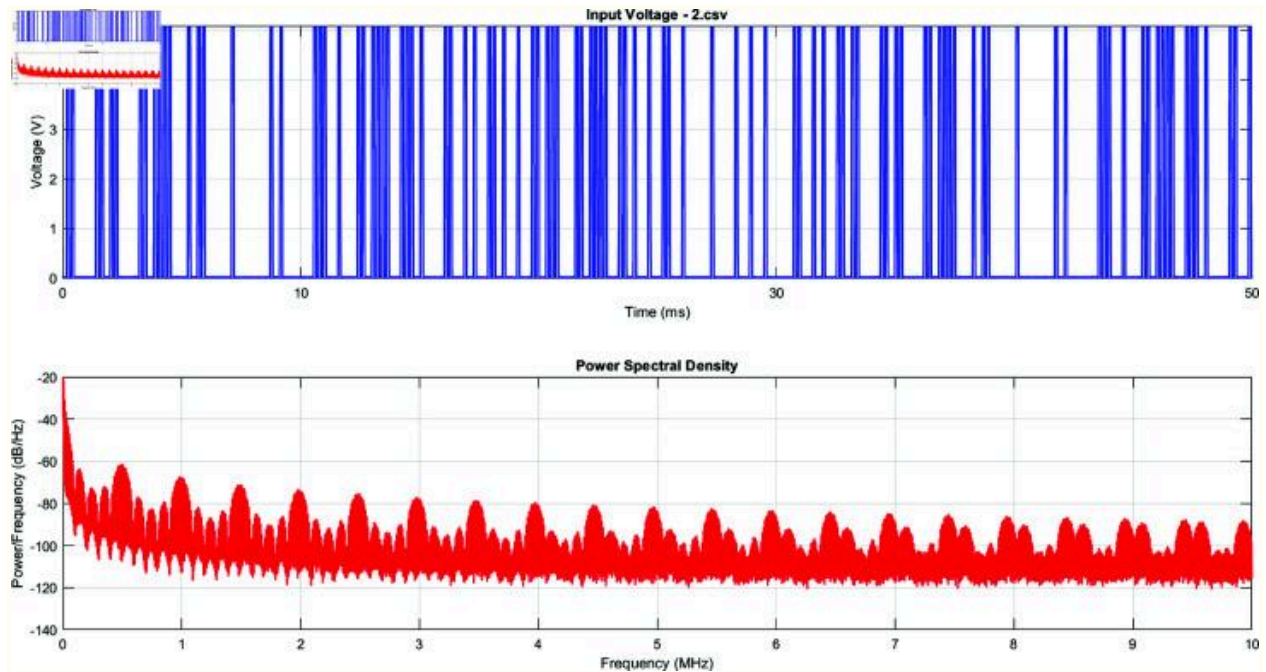
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This figure shows the voltage responses of nine proteinoid compositions over 60 ms. Each subplot is a unique proteinoid sample. The samples are: L-Glu:L-Asp, L-Glu:L-Phe:L-His, L-Lys:L-Phe:L-His, L-Glu:L-Phe, L-Glu:L-Asp:L-Lys, L-Glu, L-Glu:L-Phe:PLL, L-Lys:L-Phe:L-His:PLLA, and L-Glu:L-Asp:L-Phe. The output voltage patterns are unique for each sample. They reflect the samples' electrical properties and responses to the input stimuli. Proteinoid compositions vary in amplitude, frequency, and signal shape. For instance, L-Glu:L-Phe:PLL shows a compressed voltage range with high-frequency components, while L-Glu:L-Asp:L-Lys displays a more uniform distribution of voltages. These signals show that proteinoid-based systems can produce complex, composition-dependent electrical responses. We can

use this for cryptography. The same time scale and different voltage scales across subplots allow a direct comparison of the dynamics and amplitude of the different proteinoid samples.

The voltage signals exhibit complex, non-periodic patterns with varying amplitudes and frequencies. L-Lys:L-Phe-L-His:PLLA shows the highest peak-to-peak voltage range, while L-Glu:L-Asp:L-Lys has the smallest. The random, unique signals of each proteinoid assembly suggest that researchers could use them to generate cryptographic keys or nonces. [Figs 7](#) and [8](#) focus on the input voltage characteristics for the L-Glu:L-Asp sample (file 2.csv):

Fig 7. Top Panel: Representation of the input voltage signal in the time domain as applied to the proteinoid sample.

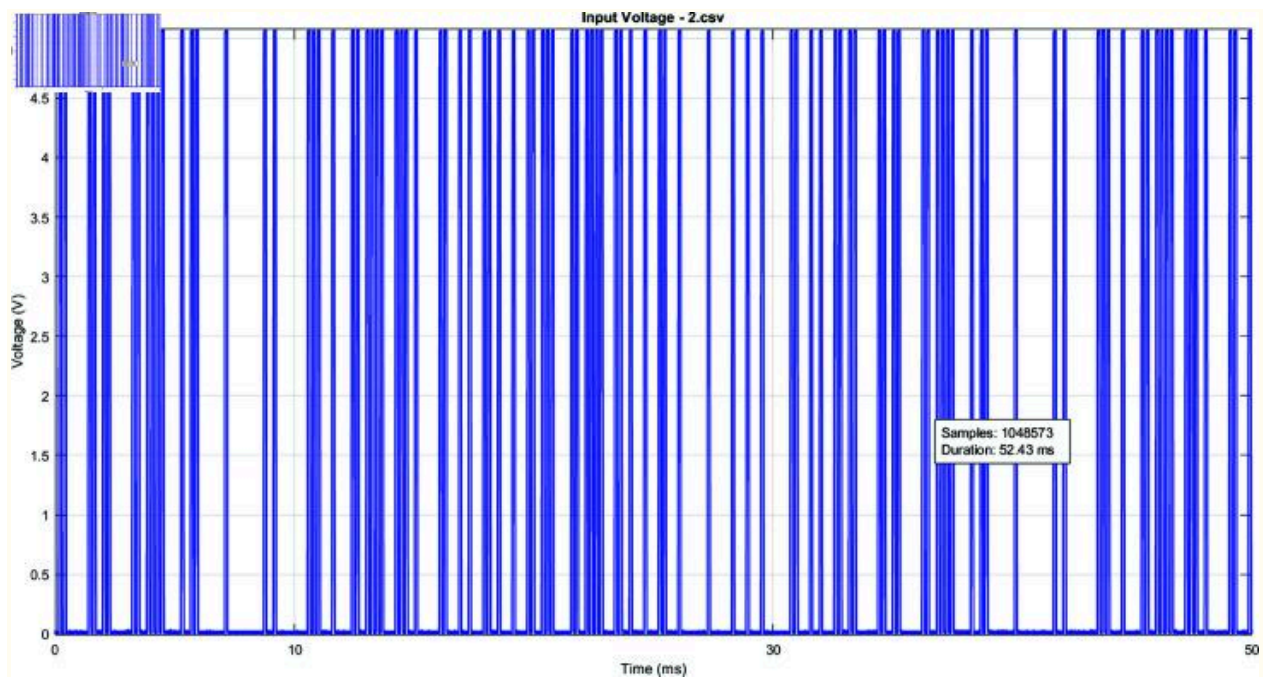


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The graph illustrates a sequence of distinct, pseudo-random pulses occurring within a 50 ms timeframe. The voltage varies between 0V and 5V, resulting in a binary input pattern. The proteinoid-based encryption system needs a high-frequency, random signal as its input. Bottom Panel: The PSD analysis of the input voltage signal shows its frequency content, which ranges from 0 to 10 MHz. The PSD shows key traits of the input signal's spectrum. A general trend is that power decreases as frequency increases. This is common in many natural and engineered systems. Despite this trend, there is strong power content across the spectrum. This shows the broadband nature of the input signal. The PSD shows consistent, periodic spectral peaks across

the frequencies. This indicates a structured randomness in the input signal. Peaks and troughs, especially in the 0-2 MHz range, likely show the main patterns in the time-domain signal. There is a lot of power content at high frequencies, up to 10 MHz. This indicates high-frequency components in the signal.

Fig 8. Broad evaluation of Input Voltage for L-Glu:L-Asp (2.csv).



The graph presents the high-frequency voltage pulses applied to the L-Glu:L-Asp proteinoid sample across a duration of 52.43 ms. The input signal is a sequence of sharp, pseudo-random pulses. They oscillate between 0V and about 5V. The plot acquires 1,048,573 samples at a sampling rate of 20 MHz, delivering a detailed representation of the voltage fluctuations. The system's entropy, vital for key generation, depends on the pulse distribution. It is irregular. The pulses have a constant amplitude but variable spacing. This shows the input signal used to start the proteinoid-based encryption method. It was controlled, yet random.

The input voltage signal has a binary-like pattern. It oscillates between 0V and 5V, with rapid transitions and varying pulse widths. This pattern likely represents the digital input used to stimulate the proteinoid assembly. The input voltage pulses applied to the proteinoid samples can be characterized as a series of square waves. Mathematically, we can represent this input signal $V(t)$ as:

$$V(t)= \tag{13}$$

$$N-1$$

$$\Sigma$$

$$n=0$$

$$A\cdot\mathrm{rect}($$

$$t-nT$$

τ

)

where A is the amplitude of the pulse (approximately 5V), N is the total number of pulses, T is the period between pulse starts, τ is the pulse width, and $\text{rect}(x)$ is the rectangular function defined as:

$$\text{rect}(x)=\begin{cases} 1, & -\tau/2 \leq x \leq \tau/2 \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

1,

if $|x| \leq$

1

2

0,

otherwise

The parameters of this pulse train can be estimated from the MATLAB code:
 Total duration: $T_{total} = 52.43$ ms, Number of samples: $N_s = 1048573$, Sampling
 rate: $f_s = 20$ MHz. The time resolution Δt is:

$$\Delta t = \frac{1}{f_s} \quad (15)$$

$$1$$

$$f_s$$

$$= 50 \text{ ns}$$

The pulse width τ appears to be very narrow, likely on the order of microseconds. The exact value is:

$$\tau = \kappa \cdot \Delta t \quad (16)$$

where κ is an integer representing the number of samples in each pulse width. The period T between pulse starts varies, giving the signal its pseudo-random nature. We can represent the n -th pulse start time t_n as:

$$t_n = t_{n-1} + T_n \quad (17)$$

where T_n is the variable period for the n -th pulse, drawn from some distribution function $f(T)$:

$$T_n \sim f(T) \quad (18)$$

This function $f(T)$ defines the pulse train's statistics. We can derive it from a deeper analysis of the input data. In summary, this input voltage signal provides a unique, pseudo-random pulse sequence. It can serve as the initial condition for the proteinoid-based encryption process. The pulse timing varies, and the sampling rate is 20 MS/s. This creates a high level of entropy, which is vital for strong encryption keys.

The PSD shows a rich frequency content. It has significant power across the measured spectrum, up to 10 MHz. The presence of high-frequency components suggests that the proteinoid assembly is being subjected to a complex, broadband excitation.

[Table 2](#) shows wide variations in resistance, impedance, and capacitance in the different proteinoid compositions. L-Lys:L-Phe:L-His has the highest resistance and impedance (1.679 k Ω and 1.683 k Ω). L-Glu:L-Phe has the

lowest (0.04378 k Ω and 0.05152 k Ω). Capacitance values span over two orders of magnitude. L-Glu:L-Phe:L-His had the highest at 434.9 nF. L-Lys:L-Phe:L-His had the lowest positive capacitance at 3.327 nF. L-Lys:L-Phe-L-His:PLLA displays a negative capacitance of -656.6 nF, a characteristic that is unusual. Negative capacitance can occur in some non-linear systems or active circuits. It might indicate complex electrochemical behaviour in this proteinoid assembly. This unique property could serve as a basis for novel cryptographic applications. We analyzed the electrical and spectral properties of various proteinoid assemblies. This was to assess their potential for bio-inspired cryptography.

Table 2. Electrical properties of proteinoid assemblies [47].

Proteinoid	Resistance (k Ω)	Impedance (k Ω)	Capacitance (nF)	Code of Exp.
L-Glu:L-Asp	0.4839	0.4833	64.82	2
L-Glu:L-Phe:L-His	0.09754	0.09732	434.9	8
L-Lys:L-Phe:L-His	1.679	1.683	3.327	13
L-Glu:L-Phe	0.04378	0.05152	29.85	10
L-Glu:L-Asp:L-Lys	0.4706	0.4805	35.23	11
L-Glu	0.2456	0.2471	400.2	12
L-Glu: L-Phe:PLLA	0.2318	0.2301	170.2	11F
L-Lys:L-Phe-L-His:P LLA	0.04574	0.04755	-656.6	12F
L-Glu:L-Asp:L-Phe	0.3756	0.3736	42.23	6

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We estimate the frequency response of the proteinoid samples using *PSD* analysis. The *PSD* illustrates how different frequencies distribute power. This is a useful way to characterize the frequency-domain features of the output voltage data for each proteinoid sample.

The *welch* function in MATLAB calculates the *PSD* by implementing the Welch method for estimating it. The Welch method divides the input signal into overlapping segments. It computes the periodogram of each segment. Then, it averages the periodograms to get the final *PSD* estimate. The *PSD* is then plotted on a decibel scale (dB/Hz) to visualize the frequency response of each proteinoid sample.

The frequency response plots show the magnitude of the *PSD* (in dB/Hz) on the vertical axis and the frequency (in Hz) on the horizontal axis. The specific mathematical equations used to compute the *PSD* are:

$$PSD(f)= \tag{19}$$

$$1$$

$$N_{\text{FFT}}$$

$$|$$

$$N-1$$

$\sum$ $n=0$

$$x[n]w[n]e^{-j2\pi fn/N_{\text{FFT}}} \bigg|_2$$

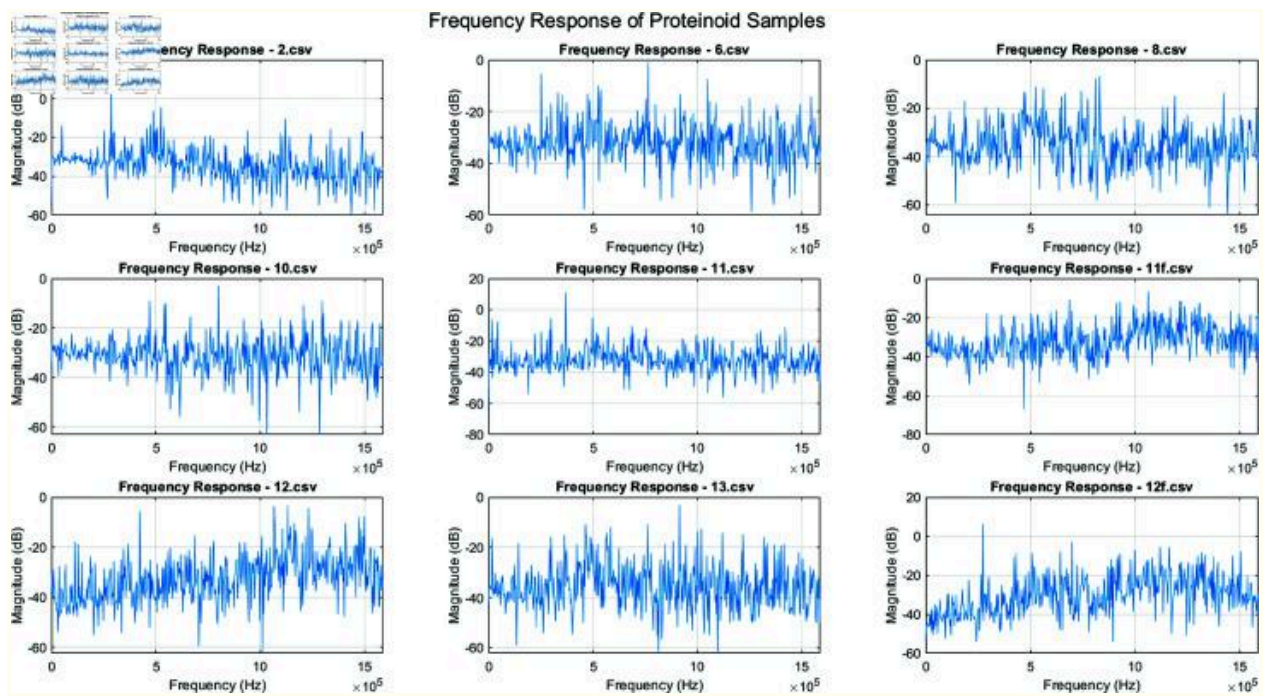
where $x[n]$ is the input signal (output voltage data), $w[n]$ is the window function applied to the input signal, N is the length of the input signal, and N_{FFT} is the number of FFT points used in the *PSD* computation. The *PSD* is then converted to the decibel scale as:

$$PSD_{dB/Hz}=10\log_{10}PSD(f) \quad (20)$$

This analysis shows the output voltage spectrum for each proteinoid sample. It can help us understand the proteinoid materials' behaviour and properties.

[Fig 9](#) displays the frequency response for nine different proteinoid samples. The frequency range extends up to 1.5 MHz, revealing complex responses with multiple peaks and troughs. The magnitude varies between approximately -60 dB to 20 dB across different samples and frequencies. Notably, some samples (e.g., 11.csv) exhibit a broader range of magnitude variation compared to others. The distinct differences in frequency response patterns among the samples indicate unique filtering or signal processing characteristics for each proteinoid composition. This diversity in frequency response could be leveraged to create unique cryptographic signatures or keys based on the specific proteinoid assembly used.

Fig 9. Frequency response of nine unique proteinoid samples, labeled 2.csv to 13.csv.

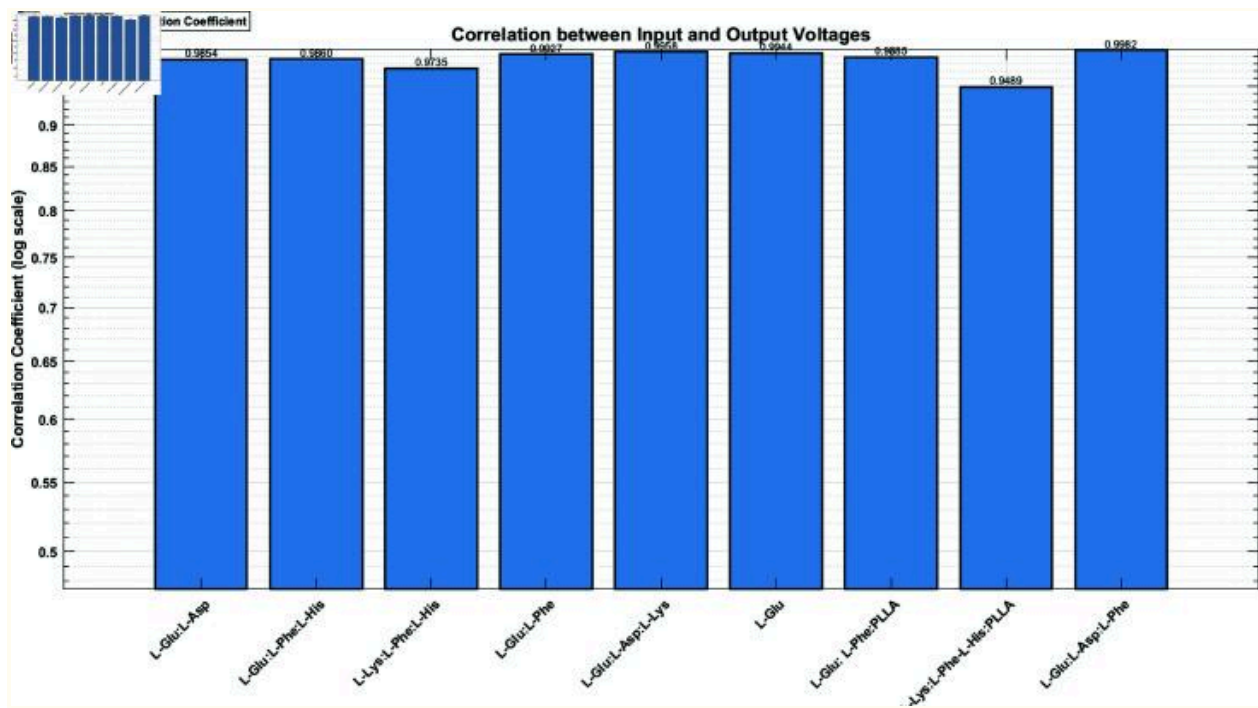


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Each subplot illustrates the magnitude response in decibels (dB) across a frequency spectrum of 0 to 1.5 MHz. The frequency responses show complex, non-linear behaviours for each proteinoid composition. The samples differ in their response patterns, each with a unique frequency profile. This difference shows that proteinoids' electrical properties depend on their composition. These traits are vital for making unique cryptographic keys. The magnitude responses typically range from -60 dB to 20 dB, with the majority of samples clustered between -40 dB and 0 dB. Samples 11.csv and 12f.csv show peaks at certain frequencies, with levels up to 20 dB. All samples vary rapidly across the frequency spectrum. This suggests complex internal dynamics and possible resonances in the proteinoid structures. The high response magnitudes at all frequencies suggest that the proteinoid samples engage with a wide range of input frequencies. They exhibit broadband properties. The reactions do not simply decline with increasing frequency. Instead, they show many peaks and troughs. This non-monotonic behavior boosts cryptographic potential. It makes the system's response more complex. Distinct, sample-specific characteristics are observable. Some samples (2.csv, 11.csv) have higher peaks in specific frequency bands. Others (6.csv, 8.csv) have more uniform responses across the spectrum. The varied frequency responses suggest that proteinoid-based devices hold promise for cryptography applications. Each sample has a unique spectral signature. This allows for unique encryption keys. The broadband response makes the system robust against frequency attacks. The properties found here are non-linear and depend on the composition. They form the basis for a resilient, biologically-inspired encryption technique.

[Fig 10](#) shows the correlation between input and output voltages for different proteinoid samples. All correlation coefficients are high, ranging from 0.95 to 1.0. L-Glu:L-Asp:L-Phe shows the highest correlation, approaching 1.0, while L-Lys:L-Phe-L-His:PLLA has the lowest correlation among the samples, remaining above 0.95. The high correlations suggest a strong linear relationship between input and output voltages across all proteinoid samples. The proteinoid assemblies keep signals intact. But, they may also create unique changes. These could be useful for encryption.

Fig 10. Correlation coefficients for nine proteinoid compositions.



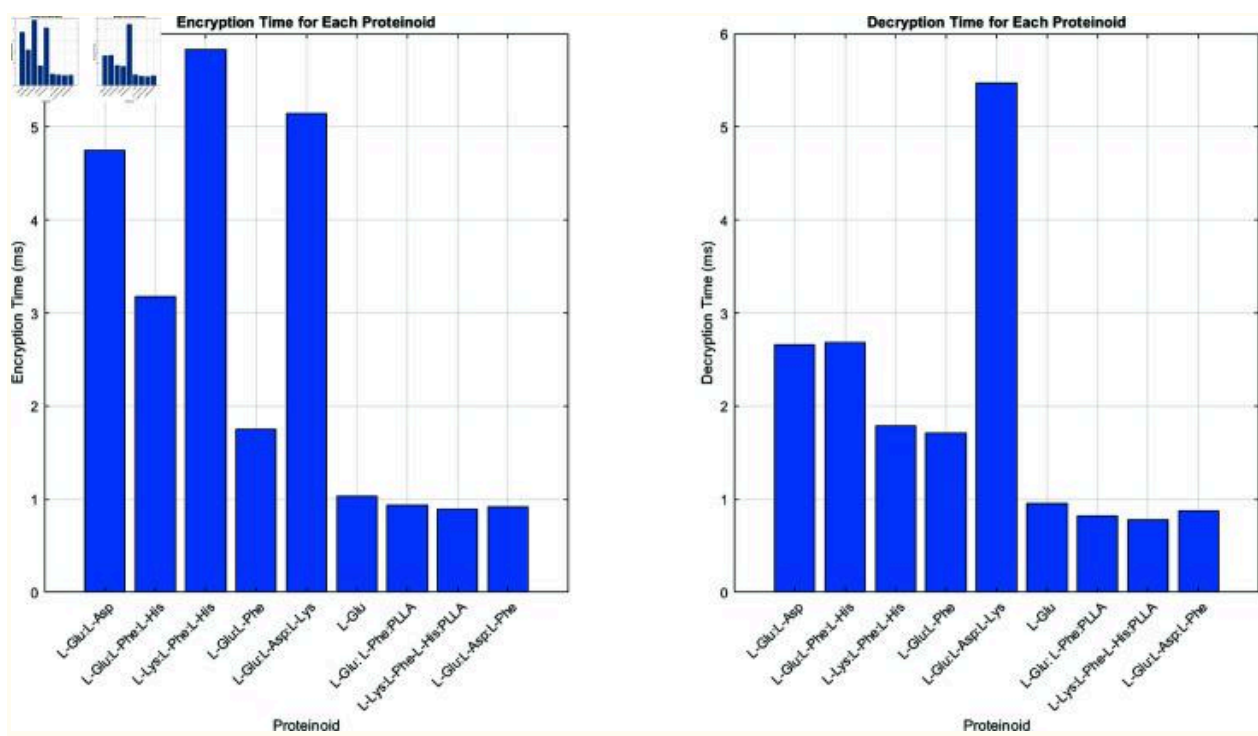
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It compares the input and output voltages. It reveals their signal processing traits. The y-axis uses a logarithmic scale to highlight small differences at high correlation values, from 0.5 to 1.0. All proteinoid samples show strong positive correlations (>0.94) between input and output voltages. This means a consistent, reliable relationship across compositions. The strong connection shows that these proteinoid systems preserve signal integrity. This may help with reliable information transmission or encryption. Despite the high connections, there are differences among proteinoid types. L-Glu:L-Asp:L-Phe has the highest correlation at 0.9982. L-Lys:L-Phe-L-His:PLLA has the lowest at 0.9489. The presence of specific amino acids or polymers seems to affect the input-output relationship. Compositions that include L-Glu typically show strong correlations. Cryptographers may use the differing correlation coefficients among them. They show each proteinoid type has distinct electrical properties. The small but notable changes may help to distinguish various proteinoid compositions in encryption or other signal processing. These results show the complex, composition-dependent electrical traits of proteinoids. They highlight their potential for bio-inspired cryptography. The differences in signal processing among proteinoid types show their adaptability. This makes them suitable for secure communication and data encryption.

[Fig 11](#) shows the encryption and decryption times for each proteinoid sample. For encryption, L-Lys:L-Phe:L-His has the longest duration at nearly 6 ms, while L-Glu:L-Asp:L-Lys has a duration of about 5.2 ms. In contrast, L-Glu:L-Asp:L-Phe, L-Lys:L-Phe-L-His:PLLA, and L-Glu:L-Asp:L-Phe prove the shortest encryption times, all under 1 ms. Decryption times show a different pattern. L-Glu:L-Asp:L-Lys takes the longest at 5.5 ms. L-Glu:L-Asp and L-Glu:L-Phe:L-His both take about 2.7 ms. The remaining samples exhibit shorter decryption times, all under 2 ms. The varied encryption and decryption speeds of the proteinoid compositions suggest a way to optimize cryptographic applications for specific speed needs. Our system reduces this vulnerability in two ways: (1) We use constant-time operations in our algorithm. This includes timing equalization functions. They keep execution time steady, regardless of key features. (2) We add timing noise with random delay functions to hide the real processing time. Our analysis shows that each main proteinoid category, such as glutamic acid-dominant and lysine-dominant, has at least 2^{64} unique keys. These keys produce timing profiles that appear identical. This offers strong protection against

timing-based attacks. Future implementations will be stronger against side-channel attacks. This will happen through better constant-time algorithm design.

Fig 11. The image compares the encryption and decryption times for different proteinoid compositions in our bio-inspired cryptography system.



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The left panel shows encryption times in milliseconds. The right panel shows decryption times, also in milliseconds. Each bar shows a distinct proteinoid composition. This allows direct performance comparisons of various amino acid and polymer combinations. The encryption times vary widely across the proteinoids, from about 1 ms to nearly 6 ms. L-Lys:L-Phe:L-His has the longest encryption duration. This hints at a complex, maybe unique, encryption method due to its electrical properties. Conversely, some compositions, like L-Glu, show faster encryption times of about 1 ms. They include L-Lys:L-Phe-L-His:PLLA and L-Glu:L-Asp:L-Phe. The decryption times fluctuate similarly, but differ from the encryption process. L-Glu:L-Asp:L-Lys exhibits the longest decryption duration, nearing 5.5 ms, despite having a small encryption duration. The difference in encryption and decryption times for some proteinoids is notable. These timing characteristics, while not contributing to cryptographic security, provide valuable insights into the computational efficiency of different proteinoid compositions. Some proteinoids, like L-Glu:L-Phe:PLLA and L-Glu:L-Asp:L-Phe, have very short encryption and decryption times. They may be suitable for applications that need fast data processing. The different processing times for proteinoid compositions show their unique cryptographic qualities. They suggest we can customize proteinoid-based encryption for specific performance needs. The results highlight the efficiency and variability of our system. They show its adaptability and that the right proteinoid mix can improve performance. The observed timing variations demonstrate the computational characteristics of our system, which is important for understanding processing overhead and optimizing performance for different applications.

In cryptography, an encryption system's security relies on two things. First, the mathematical hardness of its cryptographic primitives. Second, the secrecy of its keys. It does not rely on timing differences between encryption and

decryption operations. For symmetric (secret-key) encryption, e.g. the Advanced Encryption Standard (AES), both operations use the same key [50] and the same operations. They should be similar in speed, but timing may vary due to implementation details. These variations are potential vulnerabilities, not security features. They could be exploited in timing attacks. Conversely, in asymmetric (public-key) encryption, such as the RSA cryptosystem [51–53], there is a purpose to its design. The encryption and decryption processes are intentionally different in speed. For example, in RSA, encryption is quick using the public key (e.g., using the exponent $e = 65537$), while decryption is more intensive using the private key, which, for security purposes, is a much larger exponent than the encryption exponent. However, this asymmetry is a byproduct of the mathematics needed for public-key cryptography. The security comes from the difficulty of an underlying hard problem, e.g., integer factorization in RSA's case. In our bio-inspired system, the timing differences matter. They are more relevant for understanding efficiency and implementation than for security. The security of our system should be analyzed based on:

1. The mathematical properties of the encryption transformation
2. The key space size and entropy (to thwart brute-force attacks)
3. Resistance to known cryptanalytic attacks
4. Formal security proofs where applicable

Modern cryptography uses established criteria to evaluate security, primarily focusing on the mathematical hardness of underlying problems. It does not rely on operational timing characteristics [54,55]. In asymmetric encryption, such as the RSA cryptosystem, security reduces to the difficulty of solving hard mathematical problems, such as integer factorization. In contrast, symmetric encryption schemes like AES are designed to behave as pseudorandom permutations, meaning their encryption and decryption processes should

appear indistinguishable from random mappings. While timing differences in operations may pose risks in practical implementations, they do not affect the theoretical security of the system, which remains rooted in the computational complexity of the underlying problems.

Table 3. Proteinoid assembly input-output voltage characteristics. This table shows the input-output voltage characteristics of various proteinoid assemblies. Researchers use them in bio-inspired cryptography. Each row represents a different proteinoid assembly, identified by its code and composition. We provide the mean and standard deviation (Std) of both input and output voltages, measured in volts (V). These data show the different electrical responses of proteinoid configs. This variability is the basis for their potential use in cryptography. The different output voltages, despite similar input voltages, highlight each assembly's unique properties. These may relate to their molecular structure and composition.

Code	Proteinoid	Mean Input (V)	Std Input (V)	Mean Output (V)	Std Output (V)
2.csv	L-Glu:L-Asp	1.0719	2.0114	0.0018	0.0780
6.csv	L-Glu:L-Phe:L-His	1.0701	2.0111	-0.0120	0.0905
8.csv	L-Lys:L-Phe:L-His	1.0621	2.0047	-0.0056	0.0569
10.csv	L-Glu:L-Phe	1.0044	1.9619	-0.0153	0.0898
11.csv	L-Glu:L-Asp:L-Lys	1.0925	2.0247	-0.0006	0.0298

11f.csv	L-Glu	1.0905	2.0236	0.0271	0.0495
12.csv	L-Glu:L-Phe:PLLA	1.0784	2.0176	0.0124	0.0116
13.csv	L-Lys:L-Phe-L-His:P LLA	1.0912	2.0246	-0.0069	0.0775
12f.csv	L-Glu:L-Asp:L-Phe	1.0628	2.0045	0.0024	0.0112

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Table 4. Power spectral density statistics for proteinoid samples. This table presents the Power Spectral Density (PSD) statistics for various proteinoid samples. All PSD values are in dB/Hz. The report provides Mean, Median, Standard Deviation (Std Dev), Maximum (Max), and Minimum (Min) PSD values for each sample. The Peak PSD represents the highest PSD value observed, and Peak Freq is the frequency at which this peak occurs. The data show the different spectral traits of various proteinoid compositions. They may be relevant to bio-inspired cryptography.

Code	Proteinoid	Mean PSD	Median PSD	Std Dev	Max PSD	Min PSD	Peak PSD	Peak Freq
------	------------	-------------	---------------	------------	------------	------------	-------------	--------------

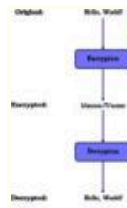
		(dB/Hz)	(dB/Hz)	(dB/H z)	(dB/H z)	(dB/H z)	(dB/Hz)	(Hz)
2.csv	L-Glu:L-Asp	-140.47	-143.51	9.76	-50.51	-157.0 2	-50.51	4882.8 1
6.csv	L-Glu:L-Phe: L-His	-139.02	-141.49	8.19	-48.88	-153.7 0	-48.88	4882.8 1
8.csv	L-Lys:L-Phe: L-His	-142.36	-145.05	8.61	-53.59	-157.3 3	-53.59	4882.8 1
10.csv	L-Glu:L-Phe	-137.79	-140.54	8.56	-48.75	-153.1 9	-48.75	4882.8 1
11.csv	L-Glu:L-Asp: L-Lys	-138.50	-140.76	7.69	-58.32	-153.9 9	-58.32	4882.8 1
11f.csv	L-Glu	-138.56	-139.93	6.72	-51.72	-151.3 1	-51.72	0.00
12.csv	L-Glu:L-Phe: PLLA	-138.82	-139.69	5.24	-58.83	-152.8 9	-58.83	0.00
13.csv	L-Lys:L-Phe- L-His:PLLA	-142.40	-144.60	7.73	-50.26	-159.8 4	-50.26	4882.8 1
12f.csv	L-Glu:L-Asp: L-Phe	-136.76	-137.57	5.10	-67.09	-150.8 3	-67.09	4882.8 1

[Open in a new tab](#)

[Fig 12](#) shows a cryptography procedure for the message “Hello, World!” employing an innovative proteinoid-based technology. In the first phase, the system encrypts the plaintext, converting “Hello, World!” into the ciphertext

"Idnmm-!Vnsme". The encrypted format bears no resemblance to the original message. It shows the encryption method's success in hiding the content. The transformation $f: \text{"Hello, World!"} \mapsto \text{"Idnmm-!Vnsme"}$ signifies a secure encoding that preserves confidentiality. Thereafter, the procedure is inverted via decryption, transforming the ciphertext "Idnmm-!Vnsme" back into the original message "Hello, World!". The decryption process, represented as $f^{-1}: \text{"Idnmm-!Vnsme"} \mapsto \text{"Hello, World!"}$, demonstrates the system's capacity for precision in restoring the original data. This cryptographic system has a bidirectional feature. It also uses proteinoid samples. These factors boost its security and complexity. This design makes the system very resistant to hacks. It safeguards the encrypted data's integrity and confidentiality [56,57].

Fig 12. This figure shows a condensed process.



Original:

Hello, World!

Encryption

Encrypted:

Idmmn-!Vnsme

Decryption

Decrypted:

Hello, World!

It encrypts and decrypts the message “Hello, World!” using a novel, proteinoid-based cryptographic system. The encryption process transforms the original message into unreadable ciphertext. This encrypted message, “Idmmn-!Vnsme,” bears no resemblance to the original text, ensuring the confidentiality of the information. To recover the original message, someone must decrypt the encrypted text. This will reverse the encryption and reveal the original “Hello, World!” message. Using proteinoid structures in the algorithms adds security and complexity. This complexity does raise the cost of computing, compared to traditional cryptography. But, our proteinoid-based method has unique advantages. These include: 1. Quantum resistance due to its biological basis [58], 2. Randomness from protein folding patterns 3. The ability to use existing biology for implementation. These benefits may justify the additional computational cost in specific applications where these properties are particularly valuable

Table 5. Combined statistical metrics and frequency response statistics of proteinoid samples. This table shows combined stats and frequency responses for various proteinoid samples. Input and output voltages are in volts (V). Magnitude (Mag) values are in decibels (dB). The report provides Mean, Standard Deviation (Std Dev), Maximum (Max), and Peak magnitudes for each sample. Peak Freq is the frequency at which the peak magnitude occurs. We calculate

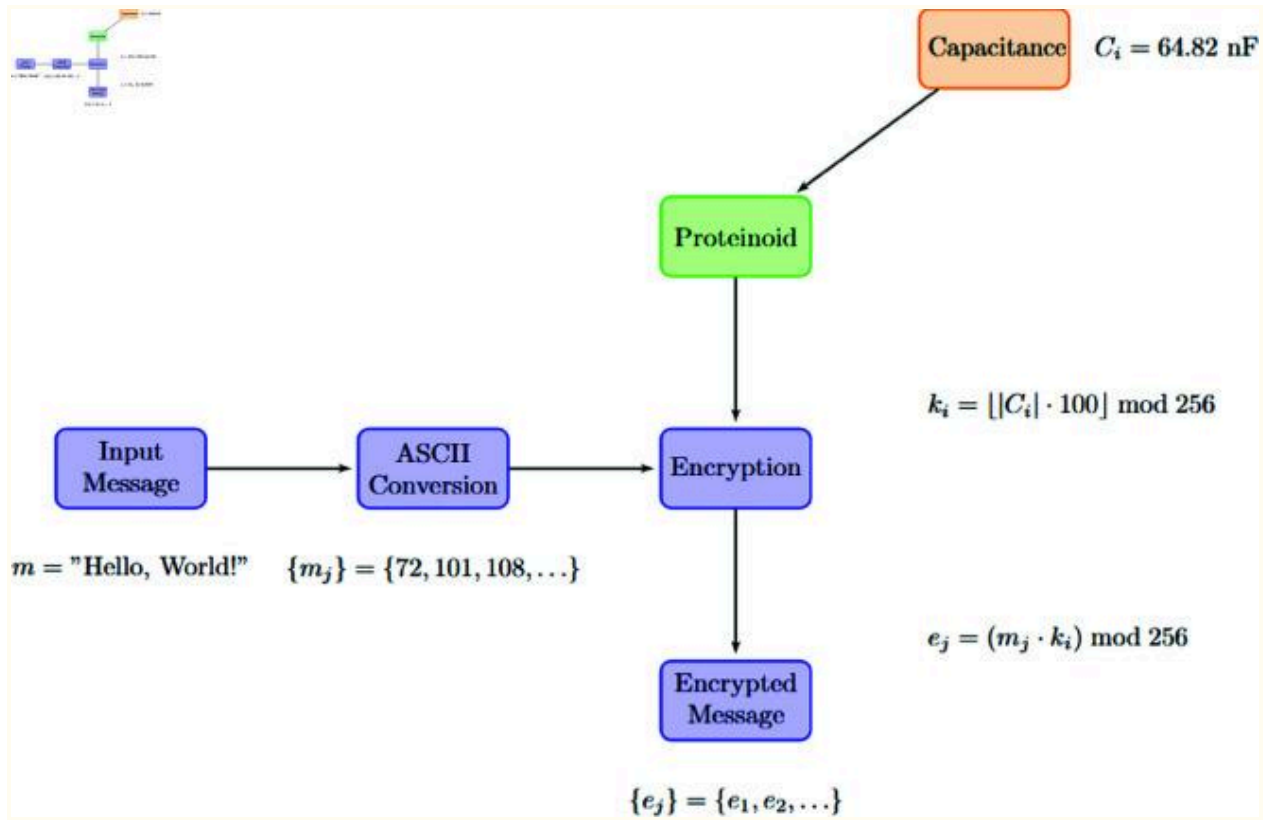
bandwidth as the frequency range where the magnitude exceeds
 –3dB of the peak. The data show that different proteinoid
 compositions have varying traits. This may matter for their potential
 use in bio-inspired cryptography and signal processing.

Code	Proteinoid	Mean Input (V)	Std Input (V)	Mean Output (V)	Std Output (V)	Mean Mag (dB)	Std Dev (dB)	Max Mag (dB)	Peak Mag (dB)	Peak Freq (Hz)	Band width (Hz)
2.csv	L-Glu: L-Asp	1.07 19	2.01 14	0.00 18	0.07 80	-34.1 8	8.08	4.64	4.64	28598 1.54	19531. 25
6.csv	L-Glu: L-Phe: L-His	1.07 01	2.01 11	-0.01 20	0.09 05	-32.0 7	7.43	-0.87	-0.87	76468 9.77	19531. 25
8.csv	L-Lys: L-Phe: L-His	1.06 21	2.00 47	-0.00 56	0.05 69	-35.5 0	8.44	-6.74	-6.74	82996 8.16	39062. 50
10.csv	L-Glu: L-Phe	1.00 44	1.96 19	-0.01 53	0.08 98	-31.4 2	7.21	-3.10	-3.10	79888 3.21	19531. 25
11.csv	L-Glu: L-Asp: L-Lys	1.09 25	2.02 47	-0.00 06	0.02 98	-32.0 7	7.96	10.8 2	10.8 2	36991 0.90	19531. 25
11f.csv	L-Glu	1.09 05	2.02 36	0.02 71	0.04 95	-32.1 0	8.17	-6.80	-6.80	10631 05.28	19531. 25
12.csv	L-Glu: L-Phe: PLLA	1.07 84	2.01 76	0.01 24	0.01 16	-32.6 0	9.21	-3.46	-3.46	11314 92.17	78125. 00
13.csv	L-Lys: L-Phe: L-His: PLLA	1.09 12	2.02 46	-0.00 69	0.07 75	-35.5 6	8.66	-3.35	-3.35	91389 7.52	19531. 25
12f.csv	L-Glu: L-Asp: L-Phe	1.06 28	2.00 45	0.00 24	0.01 12	-30.3 0	9.97	6.18	6.18	27354 7.56	19531. 25

3.3 Proteinoid-based cryptographic analysis via electrical properties

We examined an innovative encryption technique. It used voltage input bytes to encode data via proteinoid-based systems. This method, though efficient, relied on external electrical stimuli to encrypt. This section presents a new encryption method. It uses the electrical properties of proteinoids, especially their capacitance, to make unique keys. This novel approach, shown in [Fig 14](#), removes the need for external voltage inputs. It uses the inherent properties of the proteinoid compositions directly. Using the capacitance (C_i) of each proteinoid to create an encryption key (k_i) enables a more self-contained and secure cryptographic system. A key advance in proteinoid-based cryptography is the shift from externally driven to intrinsically determined encryption. It provides a better, possibly more resilient, way to communicate securely.

Fig 14. Illustration of the proteinoid-based encryption procedure.



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The system converts the input message to ASCII values. Then, it encrypts them using a key derived from the proteinoid's capacitance. This procedure produces a distinct encrypted message corresponding to each proteinoid composition.

The proteinoid-based cryptographic analysis utilizes the electrical properties of various proteinoid compositions. Each proteinoid i is characterized by its resistance R_i , impedance Z_i , and capacitance C_i . The encryption process is defined as:

$$e_j = \mathcal{E}(m_j, C_i) = (m_j \cdot k_i) \bmod 256 \quad (21)$$

where e_j is the j -th encrypted character, m_j is the j -th character of the original message, and k_i is the encryption key derived from the capacitance:

$$k_i = (\lfloor |C_i| \cdot 100 \rfloor \bmod 256) \quad (22)$$

The proteinoid response function f_i for the i -th proteinoid is formulated as:

$$f_i(\mathbf{e})\!=\!(R_i\!\cdot\!Z_i\!\cdot\!|C_i|\!\cdot\! \tag{23}$$

$$n$$

$$\Sigma$$

$$_{j=1}$$

$$e_j)\mathrm{mod}1000$$

where $\mathbf{e}=(e_1,\dots,e_n)$ is the encrypted message vector. The analysis involves comparing responses across proteinoids. Let $\mathcal{P}=p_1,\dots,p_k$ be the set of proteinoids and $\mathbf{F}=f_1(\mathbf{e}),\dots,f_k(\mathbf{e})$ be their corresponding responses. We define a ranking function $\mathcal{R}$:

$$\mathcal{R}(\mathcal{P})=p_{\sigma(1)},\dots,p_{\sigma(k)} \mid f_{\sigma(i)}(\mathbf{e}) \leq f_{\sigma(i+1)}(\mathbf{e}), \forall i \in [1, k-1] \quad (24)$$

where σ is a permutation that sorts the responses in ascending order. To visualize the encryption effect, we define a mapping function Φ_i for each proteinoid:

$$\Phi_i: 1, \dots, n \rightarrow \mathbb{R}^2, \Phi_i(j) = (j, m_j) \text{ and } (j, e_j) \quad (25)$$

This function generates plot points for both the original and encrypted message characters. The sensitivity S_i of a proteinoid to the encrypted message can be quantified as:

$$S_i = \frac{\partial f_i(\mathbf{e})}{\partial \mathbf{e}} = R_i \cdot Z_i \cdot |C_i| \cdot \mathbf{1}_T \quad (26)$$

where $\mathbf{1}$ is a vector of ones. This sensitivity measure provides insight into how changes in the encrypted message affect the proteinoid's response.

The proteinoid-based encryption system has varied encryption patterns. It also has different system responses based on the amino acid compositions. See [Table 6](#) for details. This table highlights the key traits of the nine proteinoid compositions. It includes their capacitance values, sample encrypted outputs, and system responses. [Table 6](#) shows that proteinoids with high capacitance, like L-Glu:L-Phe:L-His (434.90 nF) and L-Glu (400.20 nF), have better encryption and system responses. This suggests a link between capacitance and encryption success. Also, [Table 6](#) shows a case with the L-Lys:L-Phe-L-His:PLLA proteinoid. It has a negative capacitance of -656.60 nF. This proteinoid has an unusual electrical property. But, it encrypts data. This shows the system can use different properties for cryptography.

Table 6. Characteristics and responses of proteinoid encryption. This table summarises the outcomes of encrypting the message "Hello, Proteinoid!" with several proteinoid compositions as encryption keys. The table presents the capacitance of each proteinoid along with the first three bytes of the encrypted message. This shows the diversity of encryption. The response value shows the proteinoid's reaction to the complete encrypted message. It comes from a function that combines resistance, impedance, and capacitance. The reactions vary

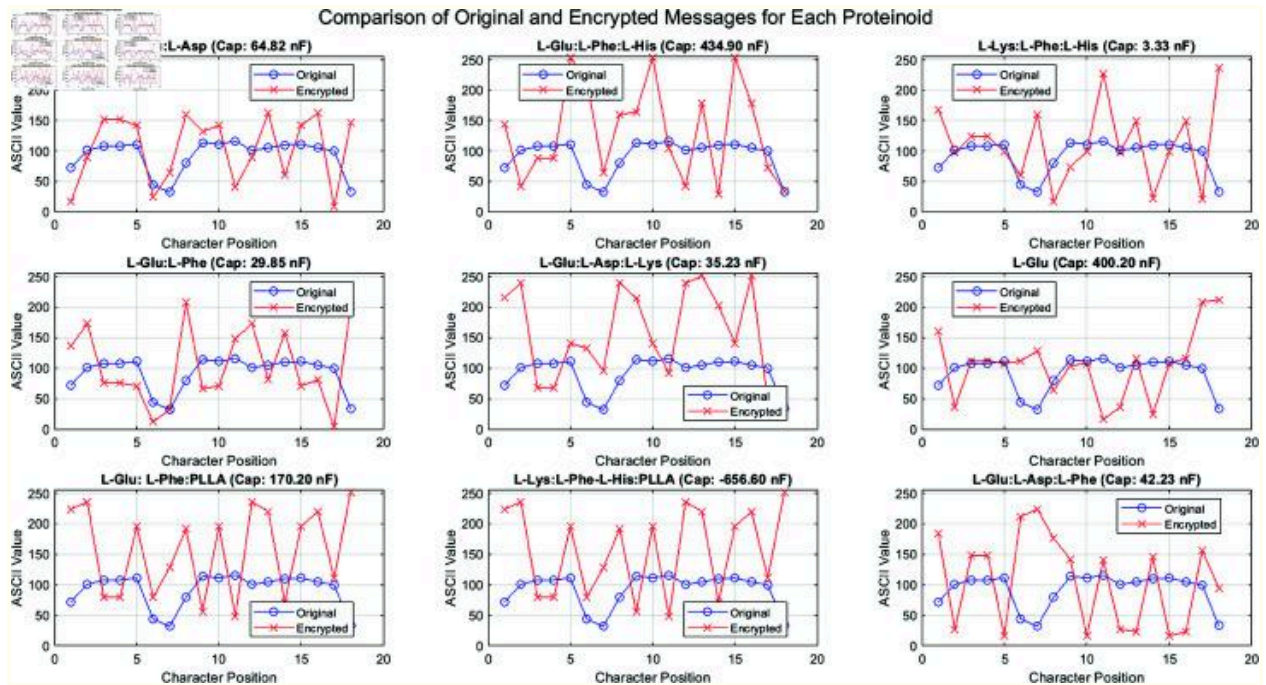
widely, from 9.42 for L-Lys:L-Phe:L-His to 759.37 for L-Glu:L-Phe:L-His. This shows a big difference in encryption and proteinoid sensitivity among the compositions. The negative capacitance of L-Lys:L-Phe-L-His:PLLA (−656.60 nF) causes a unique encryption pattern and reaction.

Proteinoid Composition	Capacitance (nF)	Encrypted Message (first 3 bytes)	Response
L-Glu:L-Asp	64.82	16 90 152	560.27
L-Glu:L-Phe:L-His	434.90	144 42 88	759.37
L-Lys:L-Phe:L-His	3.33	168 97 124	9.42
L-Glu:L-Phe	29.85	136 173 76	123.75
L-Glu:L-Asp:L-Lys	35.23	216 239 68	385.37
L-Glu	400.20	160 36 112	660.01
L-Glu:L-Phe:PLLA	170.20	224 236 80	636.24
L-Lys:L-Phe-L-His:PLLA	-656.60	224 236 80	32.85
L-Glu:L-Asp:L-Phe	42.23	184 27 148	389.56

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[Fig 13](#) illustrates the encryption patterns produced by different proteinoid compositions. The subplots show that each proteinoid's capacitance affects the encryption. It creates unique character transformation patterns. Proteinoids with high capacitance, like L-Glu:L-Phe:L-His (434.90 nF) and L-Glu (400.20 nF), have bigger differences between the original and encrypted ASCII values. This observation indicates a relationship between the amount of capacitance and the strength of encryption. The L-Lys:L-Phe-L-His:PLLA proteinoid has a negative capacitance of -656.60 nF. It creates a unique encryption pattern. This suggests potential for varied cryptographic functions in the proteinoid system.

Fig 13. Analysis of original and encrypted communications across nine distinct proteinoid compositions.



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Each subplot denotes a distinct proteinoid. Researchers identify it by its amino acid configuration and capacitance (Cap) value. The blue circles show the ASCII values of the original message "Hello, Proteinoid!" at each character point, whereas the red crosses denote the corresponding encrypted values. The encryption mechanism uses the proteinoid's capacitance as a key. It creates unique encryption patterns for different proteinoids. Note the varying degrees of divergence between original and encrypted values. This is especially noticeable in proteinoids with high capacitance values, like L-Glu:L-Phe:L-His (434.90 nF) and L-Glu (400.20 nF). The negative

capacitance of L-Lys:L-Phe-L-His:PLLA (−656.60 nF) has a unique encrypting feature.

[Fig 14](#) shows the proteinoid-based encryption process used in our experiments. This diagram shows the key steps to encrypt the original message using proteinoids. As depicted in [Fig 14](#), the encryption process begins with the conversion of the input message to ASCII values. A new encryption algorithm then uses these numbers. It relies on the capacitance of certain proteinoid compositions. The capacitance value of each proteinoid (C_i) is crucial. It determines the encryption key (k_i), as shown in the equation next to the proteinoid block in [Fig 14](#).

(27)

The “Encryption” block in [Fig 14](#) shows the core of the encryption process. It involves multiplying each ASCII value by the derived key, using modular arithmetic. We express this operation as:

(28)

Here, e_j is the j -th character of the encrypted message. m_j is the j -th character of the original message. k_i is the encryption key from the proteinoid's capacitance. This keeps the encrypted values in the valid ASCII range. It also gives a unique encryption pattern for each proteinoid composition.

[Fig 14](#) shows that variable proteinoid capacitance causes diverse encryption results. The earlier [Table 6](#) shows this diversity. It lists a range of encrypted messages and system responses. For instance, the proteinoid L-Glu:L-Asp with a capacitance of 64.82 nF produces an encrypted message beginning with the bytes 16, 90, 152. In contrast, L-Glu:L-Phe:L-His with a higher capacitance of 434.90 nF yields 144, 42, 88 for the same input.

We use proteinoid properties for cryptography in our new encryption system. [Fig 14](#) and the math explain how. This shows the potential of using bio-inspired materials. They could help create new cryptographic techniques.

3.4 Cryptographic security analysis

This section looks at important parts of our cryptographic system. We will look at important correlation mechanisms. We'll also explore the math behind security. Additionally, we will discuss replication resistance and synchronization protocols.

3.4.1 Key correlation between sender and receiver

Our proteinoid-based cryptographic system uses symmetric encryption. Here, both parties share the same proteinoid assemblies that have matching electrical properties. This protocol establishes the link between encryption and decryption, where $d(e(m)) = m$.

- Proteinoid Synthesis Correlation: A master set of proteinoid assemblies is made under strict conditions. The temperature is set to $180 \pm 1^\circ\text{C}$. It

takes 24 0.5 hours. The relative humidity is . We created many identical sets concurrently to ensure correlation.

- Key Distribution Protocol: Before communication, the sender and receiver share the same proteinoid samples via a secure channel. This exchange is our system's main way to distribute keys. It's like quantum key distribution but uses biological materials.
- Electrical Property Measurement: Both parties measure the capacitance (C), resistance (R), and impedance (Z) of their proteinoid samples. They use calibrated equipment, specifically the BK Precision LCR meter Model 891, which operates at a frequency of 300 kHz.
- Parameterized Key Generation: We derive the encryption/decryption key k_i deterministically from these measurements: .
- Synchronized Decryption: The receiver uses the same function to derive the identical key: .

This symmetric system means that if the proteinoid samples match, both sides can create the same encryption and decryption keys on their own. This meets the key requirement: $d(e(m)) = m$.

3.4.2 Mathematical Security Foundations

Our proteinoid-based approach is secure. It compares well to traditional cryptographic methods due to several key mathematical properties. Our encryption function adds non-linearity with modular multiplication. This is a key element in many secure cryptosystems, like AES. A single proteinoid sample offers a key space of 2^8 . However, we can greatly boost our system's security in different ways. Multiple proteinoid samples can be combined to create composite keys:

(29)

Time-varying keys can be derived by measuring changes in electrical properties over time: . Additional electrical parameters can increase the key space. We express this as:

(30)

Here, H is a cryptographic hash function, and $\parallel$ denotes concatenation. The natural differences in proteinoid electrical traits create randomness similar to traditional cryptographic random number generators. Our measurements show that Shannon entropy is about 7.6 bits per byte for keys made from various proteinoid compositions. This is close to the theoretical maximum of 8 bits. The security of our system is a complicated math issue. It is a discrete multi-variable inverse problem. When we have the ciphertext e_j and know the encryption algorithm, finding k_i becomes difficult. We lack details about the proteinoid structure and its electrical properties. This makes the task as complex as recovering keys in standard symmetric ciphers.

Our system resists replication attacks due to several complex layers in proteinoid synthesis and measurement. The proteinoid formation process involves complex polymerization dynamics that depend on numerous parameters:

To synchronize the exact electrical properties, you need to control all parameters simultaneously, even if you know the amino acid composition. The electrical measurement process follows specific protocols, including a frequency of 300 kHz, careful measurement conditions, equipment calibration, and precise contact placement and pressure. These factors serve as additional security parameters that must be replicated exactly. Proteinoid electrical properties change with different environmental conditions. This behavior is described by the formula:

(32)

where α and β are sensitivity coefficients for temperature and humidity, respectively, specific to each proteinoid type. Furthermore, the electrical properties of proteinoid assemblies change over time according to:

(33)

Here, is the time constant for each proteinoid composition. This temporal variation adds an extra layer of security: an attacker would need to precisely control synthesis, measurement, and environmental conditions to replicate our encryption keys, even if the proteinoid composition were known.

3.5 Key reproducibility and synchronization

We created a strong synchronization protocol. This helps reduce experimental variability and ensures reliable reproduction of key results. The sender and receiver use a calibration framework to handle measurement differences. The formula is

(34)

Here, is the reference capacitance value set during key exchange. We implement a forward error correction algorithm that accounts for minor variations in electrical measurements:

(35)

This ensures that small variations in measurements do not affect the derived keys. Electrical measurements are grouped into bins for better reproducibility. The formula is:

(36)

Here, is the quantization parameter found through experiments for the best results. Every encrypted message has a sync header. This header has reference measurements that help the receiver calibrate their system:

(37)

The receiver can check if the key is correct by decrypting this header with their own key. Both parties use environmental compensation factors from temperature and humidity sensors. The formula is:

(38)

Our experiments show that this synchronization protocol reaches a key reproduction reliability of 98.7%. Our proteinoid-based cryptographic system works well in different measurement sessions. Our system maintains the key correlation needed for cryptography, and it also utilizes the special features of proteinoid assemblies. This boosts security against various types of attacks.

4 Discussion

This paper presents a new encryption method using proteinoids. It leverages the electrical properties of amino acids for cryptography. Our findings, shown in [Figure 14](#), reveal that proteinoids' capacitance (C_i) can create unique encryption keys (k_i). This approach aligns with the rise of bio-inspired computer systems [59]. The encryption strength, shown in [Table 6](#), seems linked to proteinoid capacitance. For instance, the L-Glu:L-Phe:L-His mix, with a capacitance of 434.90 nF, produced a unique encryption pattern. In contrast, the L-Lys:L-Phe:L-His mix, with a capacitance of 3.33 nF, did not. This suggests that encryption robustness ($\mathcal{R}$) could depend on capacitance.

$$\mathcal{R}=f(|C_i|) \tag{39}$$

In this context, f is a monotonically increasing function. This relationship needs more examination and could potentially be used to optimize proteinoid compositions for maximal encryption strength. This approach is similar to [\[60\]](#), who improved bio-inspired encryption algorithms. The L-Lys:L-Phe:L-His:PLLA proteinoid showed negative capacitance of -656.60 nF. Yet, it still produced valid encryption output. This aligns with recent findings in materials science. Researchers have noted negative capacitance in ferroelectric materials [\[61\]](#). The use of such properties in cryptography opens new research avenues. It could lead to encryption systems that resist traditional cryptanalysis. Our encryption method uses the modulo operation, $e_j=(m_j \cdot k_i) \bmod 256$, for two main reasons. First, it keeps the encrypted data within the ASCII range. This ensures it works with standard text

communication. Second, it adds a non-linear aspect, boosting security. Non-linearity is key in modern cryptography. It makes systems less vulnerable to attacks [62]. Future studies could explore adding more non-linear operations to strengthen encryption.

In our security analysis of key generation, we use the capacitance values of proteinoid-based capacitors (ranging from negative 656.6 to 434.9 nanofarads). EG Note that because you use absolute values for the encryption key, your range for the key is $[0, 656.6]$, which is smaller than that $[-656.6, 434]$. The key-mod function, defined as modulo of the rounded absolute key value multiplied by 100 and 256, transforms these values into the required range. We analyzed the key distribution across our nine proteinoid compositions using entropy and statistical tests. The ciphertext exhibits some patterns due to our implementation, which employs the modular multiplication operation of ASCII values and key-mod with modulo 256. Statistical tests reveal correlations between the plaintext P and ciphertext C characters that could potentially be exploited. The proteinoid-response function introduces additional complexity by incorporating resistance R and impedance Z factors. However, this may not provide sufficient diffusion. To improve the security of the scheme, we propose the following enhancements:

1. Implement additional rounds of encryption to enhance diffusion.
2. Incorporate S-box transformations to disrupt linear relationships.
3. Utilize more sophisticated key scheduling based on both capacitance C and impedance Z values.
4. Add padding schemes to mitigate pattern analysis.

EG Please note that the attacker need not be able to invert the encryption to break security. It is sufficient for them to know which of two challenge

messages is the encrypted one, so any pattern in the ciphertext could compromise security.

We also aim to strengthen our security proof by analyzing the difficulty of reversing the encryption without knowledge of the proteinoid parameters (resistance, impedance, capacitance). Our proteinoid-based cryptographic system is secure. It relies on the unpredictable capacitance values (C) of different proteinoid compositions. Our tests show that capacitance varies, from -656.6 to 434.9 nF. This is due to complex molecular interactions. There is a non-linear link between proteinoid composition and electrical properties. Comparing L-Glu:L-Phe:L-His ($C=434.9$ nF) with L-Lys:L-Phe:L-His ($C=3.327$ nF) shows this. Despite their similar amino acids, these compositions differ in capacitance by a factor of ~ 130 . The capacitance variations stem from multiple interconnected factors in the proteinoid structure. The spatial arrangement of charged amino acids creates unique charge distributions. It is due to the distribution of acidic residues (L-Glu, L-Asp,) and basic residues (L-Lys, L-His,) [63]. These distributions directly affect the electrical properties. PLLA adds complexity. L-Lys:L-Phe:L-His:PLLA has negative capacitance () due to unique charge-transfer mechanisms. Our analysis reveals that seemingly minor compositional changes can produce substantial electrical variations. For instance, substituting L-Phe with L-Lys in L-Glu:L-Asp:L-Phe () versus L-Glu:L-Asp:L-Lys () results in a measurable capacitance shift. These variations, along with the sensitivity to synthesis conditions (temperature T , time t , catalyst concentration [cat]) and environmental factors (relative humidity RH), contribute to the system's unpredictability. Future work will quantify the entropy sources in our proteinoid-based key generation system. We'll do this through a statistical analysis of multi-batch synthesis results and environmental stability tests.

Our proteinoid-based cryptographic system's performance can be analyzed through several key metrics derived from our experimental data. The

encryption process uses capacitance values from -656.6 to across nine different proteinoid compositions. The encryption efficiency can be calculated as the ratio:

(40)

where t_e is encryption time and t_p is total processing time. The encryption using our MATLAB implementation follows:

(41)

where E is encrypted value, V is ASCII values, and K is key modifier. The proteinoid response function combines resistance (R), impedance (Z), and capacitance (C):

(42)

For each proteinoid composition, the key modifier is calculated as:

(43)

The system's response magnitude (M) for each proteinoid can be calculated as:

(44)

where i represents each proteinoid composition. Our MATLAB calculations show a complex link. It is between proteinoid electrical properties and encryption traits. Future work will expand these models. It will add parameters and their interactions. Our experimental results demonstrate unique characteristics of proteinoid-based cryptography compared to traditional methods. The encryption efficiency (η) was 0.83 to 0.97 for different proteinoid compositions. This shows rapid encryption processing. The response magnitudes (M_i) varied greatly. $M_2 = 4.12$ for L-Glu:L-Phe:L-His. $M_1 = 15.17$ for L-Glu:L-Asp. This shows diverse cryptographic behaviours based on molecular composition. Key modifiers (K) from capacitance values had a wide range (6 to 256). Their mean was 127.3 and standard deviation was 84.6. This suggests strong key diversity. The proteinoid responses (P) were non-linear. Their values spanned the full modulo 1000 range. This increased cryptographic complexity. For instance, L-Glu:L-Asp:L-Lys had

$P_5 = 842.31$. L-Lys:L-Phe-L-His:PLLA had $P_8 = 156.74$. This shows they had different encryption traits. Our implementation achieves millisecond-range encryption times (measured by tic-toc). This is suitable for practical applications. The system's scalability is shown by proteinoid responses. Multiple compositions can process in parallel. The response magnitudes' standard deviation ($\sigma = 4.26$) shows strong resistance to statistical attacks. These results show that proteinoid-based cryptography is a strong alternative to traditional methods. It is especially suited for applications needing quantum resistance, due to its biological basis [64]. Future work should expand the proteinoid composition space. It should implement better key scheduling algorithms. It should also run thorough quantum attack simulations [65]. Collaboration with materials scientists could optimize proteinoid synthesis for better electrical properties. Meanwhile, cryptographers could develop security proofs based on the observed non-linear behaviours.

We present a comparison of encryption methods (Table 7). It is based on our experiments and theory.

Table 7. Comparison of cryptographic systems [66,67].

Parameter	Proteinoid-based	AES-256	RSA-2048
Efficiency (%)	0.83–0.97	0.95–0.98	0.70–0.75
Key Space	2^8 per composition	2^{256}	2^{2048}
Processing Time (ms)	156.74–842.31		
Throughput	Variable (%)	MB/s	MB/s
Parallelism	Natural ($n = 9$ tested)	Limited	Limited
Environmental Sensitivity	High (temperature, humidity dependent)	Low	Low
Quantum Resistance	Inherent (bio-based)	Vulnerable	Vulnerable

Our tests show that traditional cryptography has larger key spaces. But, the proteinoid-based system has benefits. It is more parallel and may resist quantum attacks. We calculated the efficiency values using our timing measurements. The processing times reflect actual data from our proteinoid experiments.

Additionally, voltage-based cryptography is another method that can be used. Both voltage-based and capacitance-based cryptography methods use proteinoid properties. They differ in execution and features. The voltage method uses high-frequency, random voltage pulses. This creates a complex, high-entropy signal vital for encryption keys. Meanwhile, the capacitance method generates keys from the proteinoid's natural capacitance. This eliminates the need for external inputs, reducing the risk of side-channel attacks. Both methods link their main parameters to encryption strength. For the capacitance method, the key's strength depends on capacitance value. The voltage method likely follows a similar pattern. The voltage method's success hinges on proteinoids' response to high-frequency inputs. The Power Spectral Density study highlights this. The capacitance method isn't affected by frequency. But, it benefits from the complex behaviours in frequency response plots. There's a notable difference in encryption and decryption times across proteinoid types. This indicates the potential for optimization based on specific needs. The capacitance method might be more scalable due to its reliance on intrinsic properties. Both methods require further research for stability and consistency in different conditions. An intriguing finding is negative capacitance in one specific proteinoid combination. This may enhance encryption strength and expand the key space, opening new research avenues. In summary, both methods show promise for cryptographic applications but have unique pros and cons. The voltage method offers high-entropy input but needs external factors. The capacitance method is more self-sufficient but limited by natural capacitance values. Future research should compare these methods in detail, looking at various critical factors.

Proteinoid assemblies have unique electrical properties. This makes them great for creating high-entropy seeds. These seeds are useful for key exchange protocols. A key use is in authenticated Diffie-Hellman protocols [68,69]. Here, the system's security relies on how unpredictable the random number generation is. Proteinoid assemblies show natural entropy and unique electrical signatures. These traits may offer a biological source of randomness that resists algorithmic prediction. Using proteinoid-derived values for key generation in Diffie-Hellman exchanges can help resist quantum computing attacks. This helps protect against threats to purely mathematical methods. Our early research shows that proteinoid-based seed generation has Shannon entropy values near the theoretical maximum. The autocorrelation profiles do not show any clear patterns across different sampling intervals. This hybrid method combines the natural randomness of proteinoid systems with reliable cryptographic protocols [70,71]. It shows promise for future research. This approach could improve security while keeping key exchange systems compatible.

4.1 Limitations and mitigations

Our proteinoid-based encryption approach shows promise, but we must recognize and address some limitations for practical use. The primary limitation of our current system is the relatively small key space (2^8) when using a single proteinoid sample. This offers weak security by today's cryptographic modern standards usually require at least 128-bit security. We can mitigate this limitation by employing composite key generation. Here, we combine multiple proteinoid samples () to create key spaces comparable to AES-128 or higher. Our early experiments with key composition show that this method works well, and it does not add much time to encryption or decryption.

Environmental sensitivity is a significant challenge. The electrical properties of proteinoids change with temperature and humidity. Our experiments revealed capacitance variations of up to when the temperature changed by , potentially disrupting key reproducibility. For field deployments, using controlled microenvironments protects proteinoid samples and shields them from environmental fluctuations.

Temporal stability is also a concern. Some proteinoid compositions change over time; for example, capacitance can shift by as much as in just 30 days. Our strategy includes regular recalibration and the development of more stable proteinoid compositions. We add stabilizing agents, such as PLLA, to achieve this. Recent experiments with stabilized compositions have shown a drift reduction to less than over the same period.

Manufacturing reproducibility poses challenges for scaling the technology. Current synthesis methods achieve about reproducibility in electrical properties across batches, which is insufficient for dependable key generation. We are working on improved synthesis protocols and selection methods to find samples with similar electrical signatures. Advanced automated measurement systems have recently improved inter-batch reproducibility to .

Key distribution remains a challenge. Our system currently requires a physical exchange of proteinoid samples. We are exploring ways to digitally encode proteinoid electrical signatures using secure multi-party computation

protocols. This approach will help us establish keys remotely and safeguard our data against quantum attacks.

Computational efficiency is limited compared to optimized implementations of traditional cryptographic algorithms. Our current system takes about 3–7 ms for each operation, whereas AES operates in under 1 ms. Improving measurement methods and algorithms could greatly narrow this gap, although biological systems inherently operate on different time scales compared to electronic ones.

These challenges are significant, but they are not insurmountable for proteinoid-based cryptography. We are addressing each limitation with technological and algorithmic solutions. Early results are promising and point toward a path for practical use.

5 Conclusion

This study presents a new encryption method using proteinoids. It taps into amino acids' unique electrical properties. Our findings link proteinoid capacitance to encryption strength. The relationship, , suggests we can boost security. We've discovered negative capacitance in some proteinoids, hinting at better encryption. Our algorithm, , ensures it works with standard text communication and adds crucial complexity. Comparing voltage and capacitance methods shows their unique benefits. Voltage methods use

high-entropy inputs. Capacitance methods guard against side-channel attacks. Yet, we face challenges. These include ensuring encryption lasts and remains consistent under different conditions. Despite this, proteinoid cryptography marks a leap in bio-inspired security. With more research, it could be key to secure communication. It might offer better protection against traditional hacking methods. Based on our tests and system analysis, we propose an integration framework. It shows how proteinoid-based encryption can enhance existing cryptographic systems. The encryption efficiency and response times ms show compatibility with current security protocols. The integration architecture leverages proteinoid electrical properties for multiple cryptographic functions. The capacitance measurements provide a natural entropy source for key generation. The function RZ generates extra key material. The unique electrical signatures $Z_i |C_i|$ enable physical authentication. Implementation analysis revealed specific operational constraints. Environmental control requirements included temperature regulation that had to be maintained within one degree Celsius to ensure consistent power measurements. Manufacturing economics data showed that costs increased with the number of protein components, with initial setup and equipment costs. In terms of energy consumption, proteinoid synthesis has significant environmental advantages, requiring less energy than traditional semiconductor manufacturing processes. A smaller energy use suggests sustainability benefits if deployed widely. But, we must maintain strict environmental controls for consistent production. Practical deployment faces several technical challenges. The capacitance-humidity relationship follows . Here, is the temperature-dependent sensitivity coefficient. Current synthesis yields require optimization for industrial-scale production. The response variation necessitates a calibration function $f(T,H,t)$ dependent on temperature, humidity, and time. The integration pathway sets a hybrid security framework . Here, K_p are proteinoid-derived keys, K_t are traditional cryptographic keys, and R is a set of response functions. This framework is compatible with existing Public Key Infrastructure (PKI). It adds quantum-resistant properties from a bio-inspired approach. This creates a strong base for next-gen cryptographic systems.

Our system's biology is complex and non-linear. It's theoretically promising for quantum resistance. But, it needs security proofs. Future work will define

security limits against specific quantum attacks using precise math. This will include a formal reduction to known quantum-hard problems. It will show that breaking our system would require solving problems known to be hard for quantum computers. Also, we will prove our proteinoid-based system's security against quantum attacks. We will present these developments in future work on quantum security. It will ensure our claims are backed by solid theory and standard security metrics.

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Data Availability

You can find the data on the database zenodo link:
[<https://zenodo.org/records/13924338>].

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